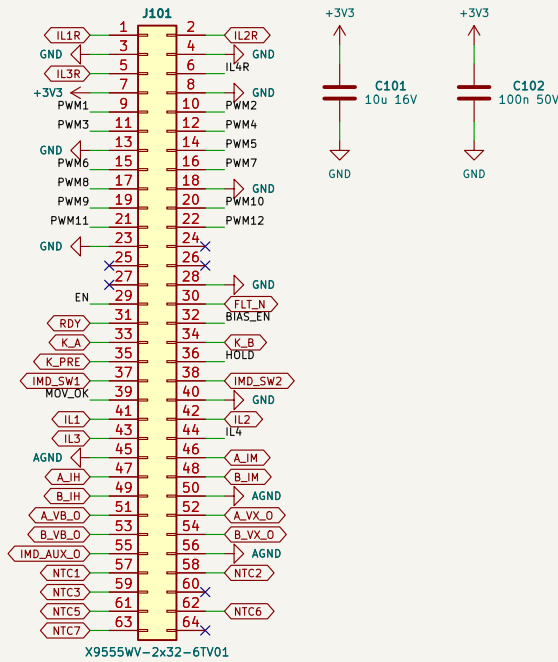


	1	2	3	4	5	6	7	8
A	PV–PWR power board							
B	Rev A2 2026–10–05 Schematics authored by Chinmoy Bhuyan PV–P75: 3 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)							
C	01 PC connector, command buffers, status interfaces.PC (2x32) to the control board, 3,3 → 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points File: sheets/01_pc.kicad_sch 02 Phase 1 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/02_ph1_power.kicad_sch 03 Phase 1 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/03_ph1_gda.kicad_sch 04 Phase 1 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/04_ph1_gdb.kicad_sch							
D	05 Phase 2 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/05_ph2_power.kicad_sch 06 Phase 2 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/06_ph2_gda.kicad_sch 07 Phase 2 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/07_ph2_gdb.kicad_sch 08 Phase 3 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/08_ph3_power.kicad_sch							
E	09 Phase 3 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/09_ph3_gda.kicad_sch 10 Phase 3 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/10_ph3_gdb.kicad_sch 11 Port A power path (pv) Varistor Y + monitor File: sheets/11_leanA.power.kicad_sch 12 Port A sensing and interlocks References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks File: sheets/12_leanA_ctrl.kicad_sch							
F	13 Port A coil drives Contactor: interlock logic, fail–safe high side, economiser; passive bank bleeder File: sheets/13_leanA_drive.kicad_sch 13 Port B power path (battery) Varistor Y + monitor, aR fuses File: sheets/13_leanB.power.kicad_sch 14 Port B sensing and interlocks References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks File: sheets/14_leanB_ctrl.kicad_sch 15 Port B coil drives Contactor + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder File: sheets/15_leanB_drive.kicad_sch							
G	15 Terminals, EMI filter, insulation test CAD CEM1015 no resonance Q using one RDT 8 series resistor number on each build (output 24 V (PV–PWR) design checked kV LVD at 1000 Hz 300 ns, 15 nV, GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board domains: LIVE (BUS– = GND = AGND), GD_PxSy gate islands (functional: NSI6651, T_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX75 from force transformer (AUXJ-BELV).CC output to PDU). Shared heatsink: Tj 107 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 19.8 W (economiser) <= aux live rating 30.0 W; module aux 62 W. Full fan speed is only guaranteed while the converter is switching. File: sheets/15_terminals_jmd.kicad_sch 16 Live 5 V / 3.3 V, SELV connector Live 23.62–24.68 V (FB); SELV 21.4–30.1 V over the load table, gates off 21.4–27.7 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low File: sheets/16_supplies.kicad_sch 17 AUX75 input taps, OR, film One tap per port (+ pole, terminal side), return = BUS–; 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 35LHV (AUX75 film forced windings (AUXJ-BELV).CC output to PDU). File: sheets/17_aux75_input.kicad_sch 18 AUX75 start–up, brown–in, OV, timer As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 39–100 ms File: sheets/18_aux75_startup.kicad_sch							
H	19 AUX75 PWM, SiC, clamp, AUX–T1 UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 50:9:10; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding File: sheets/19_pwm.kicad_sch 20 AUX75 live 24 V, SELV 24 V, status Live 23.62–24.68 V (FB); SELV 21.4–30.1 V over the load table, gates off 21.4–27.7 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low File: sheets/20_status.kicad_sch							
I	21 PCB layout, assembly, testing PCB layout, assembly, testing File: sheets/21_layout.kicad_sch 22 Bill of Materials (BOM) Bill of Materials (BOM) File: sheets/22_bom.kicad_sch							
J	23 Assembly instructions Assembly instructions File: sheets/23_instructions.kicad_sch 24 Test procedures Test procedures File: sheets/24_test.kicad_sch							
K	25 Safety analysis Safety analysis File: sheets/25_safety.kicad_sch 26 Compliance documentation Compliance documentation File: sheets/26_compliance.kicad_sch							
L	27 Change log Change log File: sheets/27_change_log.kicad_sch 28 Revision							

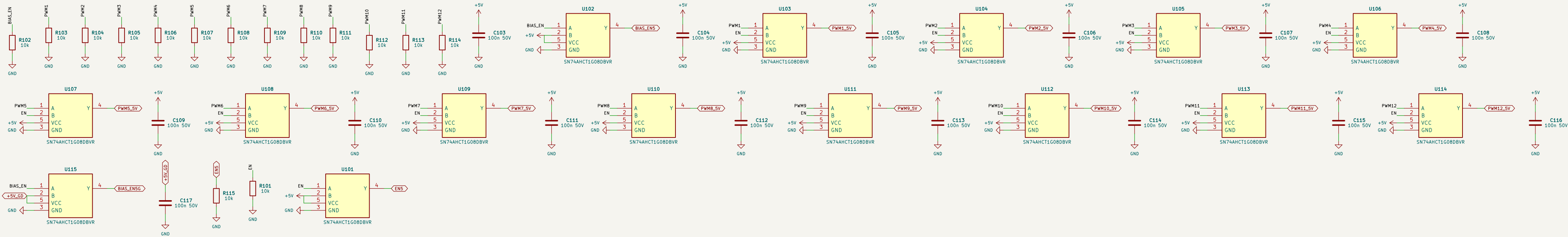
File: sheets (20 rows) - outputs: bind, ext

PC CONNECTOR (INTERFACES.PC), LIVE



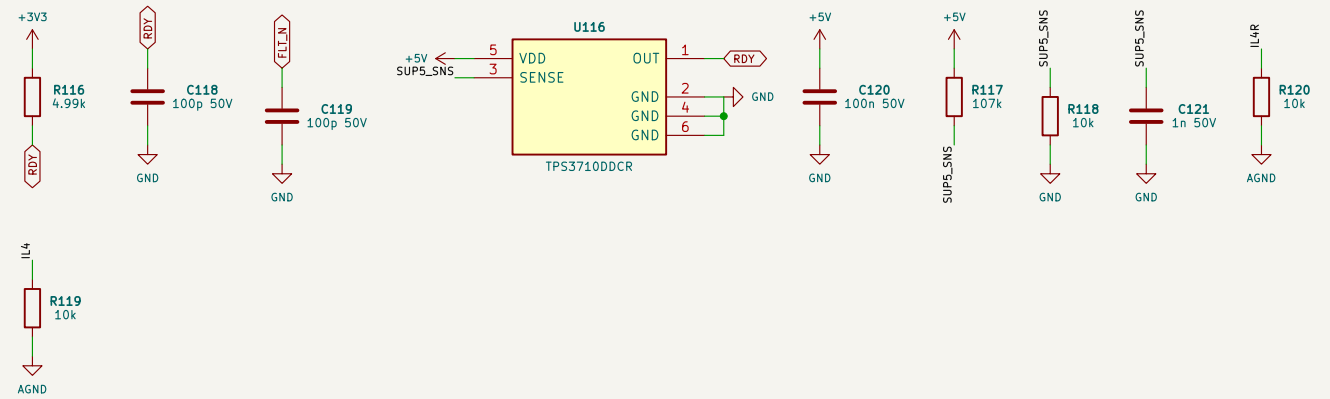
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2; the former +24V / +5V pins carry the sensor references IL1R-IL4R). Commands are 3.3 V logic after the control board's trip latch; every command input is pulled low here (open = off).
PV-P75: PWM13-16, NTC4 and NTC6 not connected; IL4 and IL4R pulled to AGND (0 V = no phase 4).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



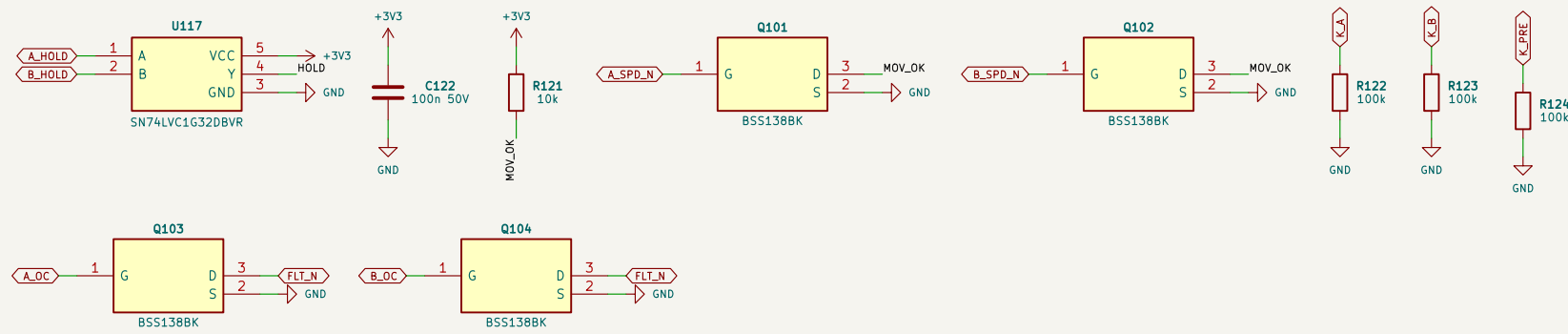
NSI6651 inputs are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48-0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



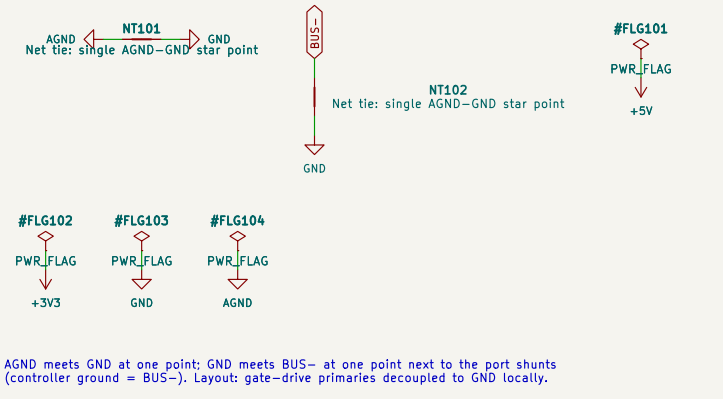
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
+5V supervisor TPS3710 107k/10k; RDY low below 4.52-4.69 V (AHC2 VCC <= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



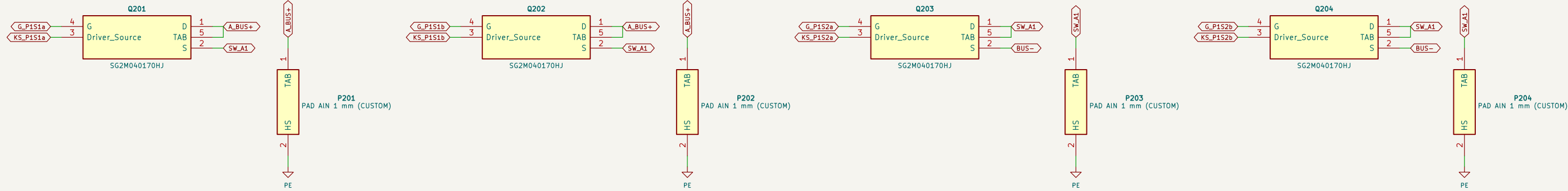
HOLD = A_HOLD OR B_HOLD ([Lport] >= 0.97-1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



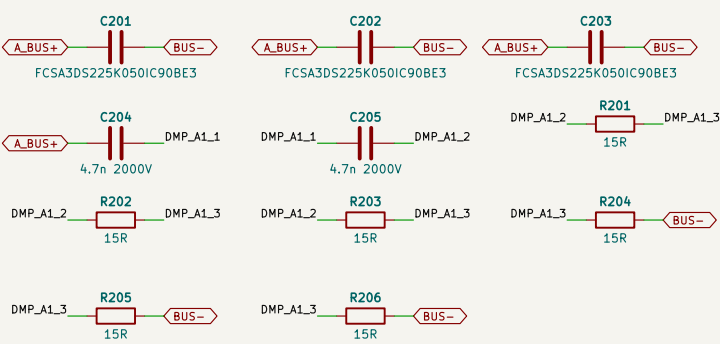
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-); Layout: gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



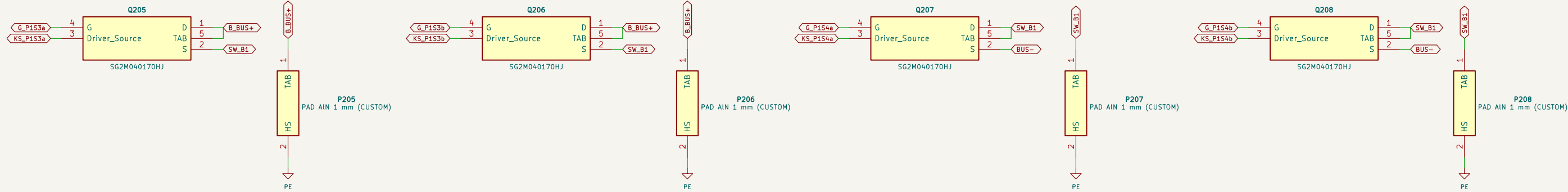
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



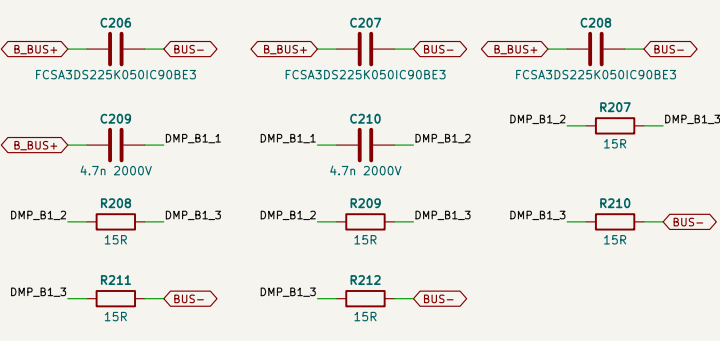
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



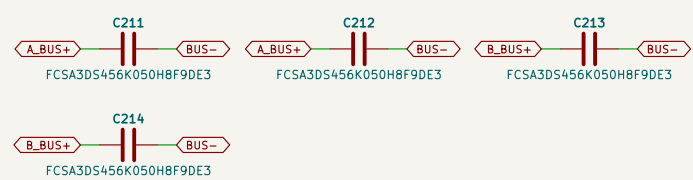
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



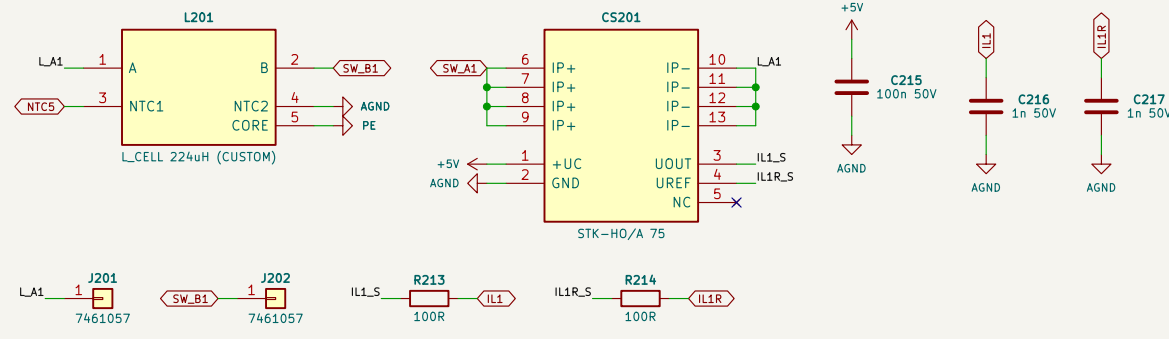
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



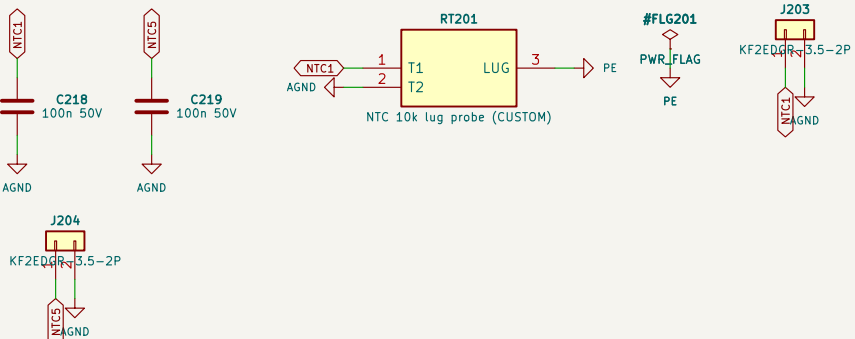
2 x FCSA3D5456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs; the banks of all phases form the port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



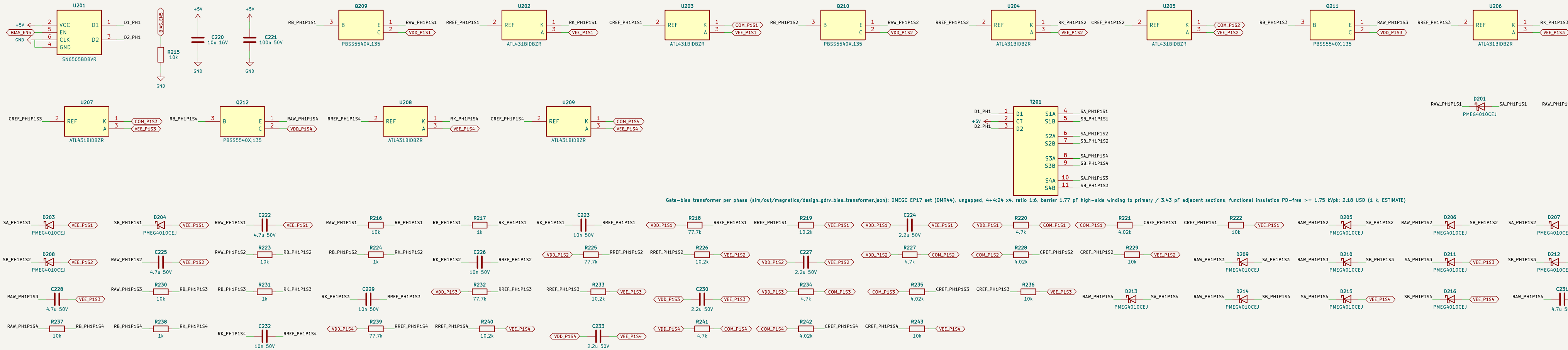
Lead A from SW_A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out + = SW_A -> SW_B) to stud LA1; lead B on stud SW_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A; 100R +/- 1 mF at the source. IL1R = the sensor's own Uref (pin 4). 100R +/- 1 mF PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



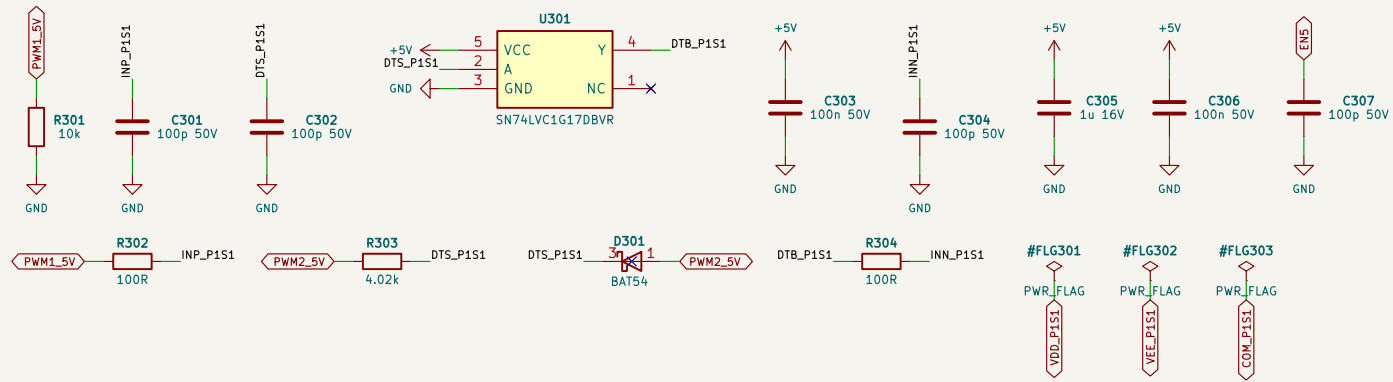
NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10). NTC5 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

PH1: GATE BIAS (PER PHASE)



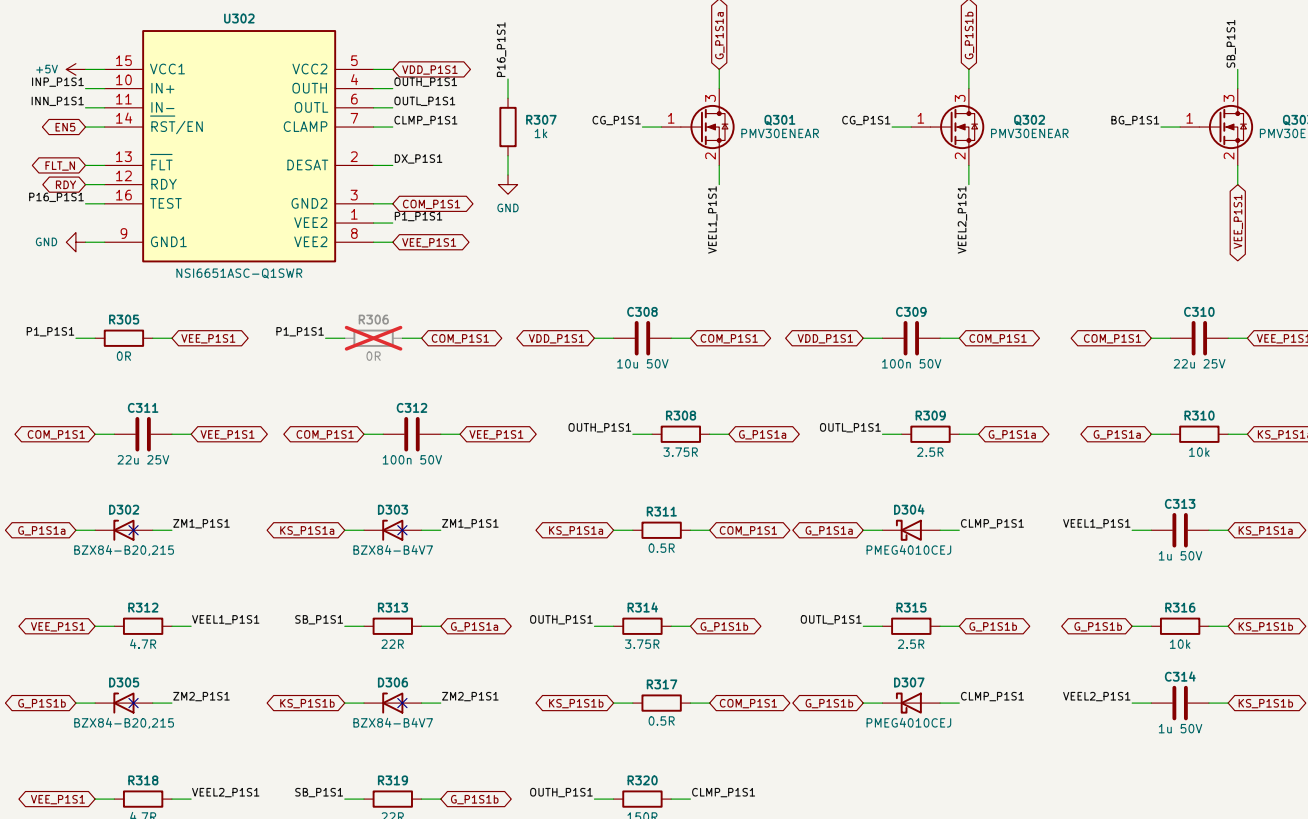
SN6505B push-pull 420 kHz from +5V. BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S1: LOGIC SIDE



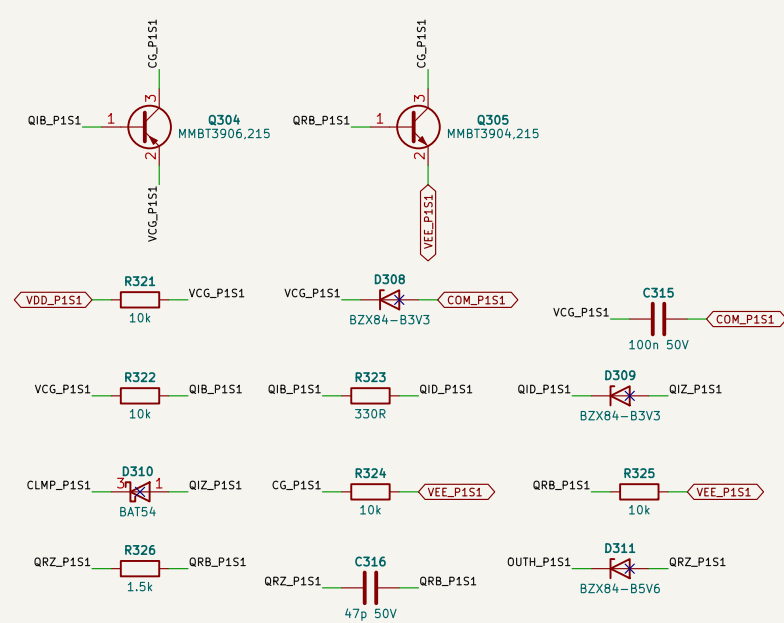
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S1: DRIVER AND 2 GATES



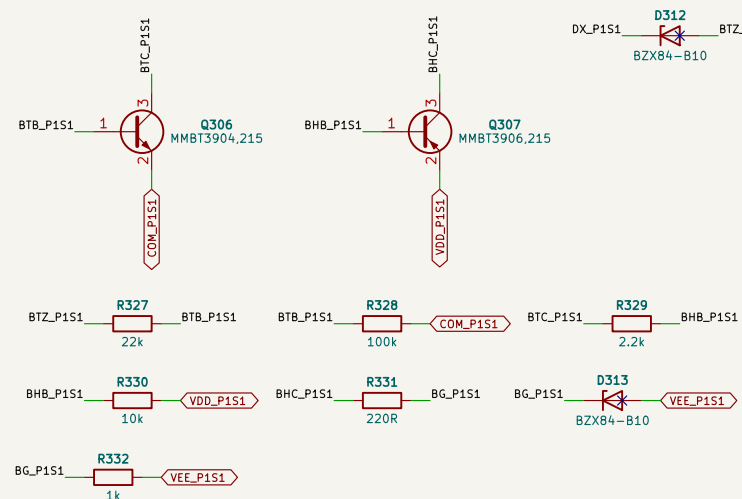
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S1: MILLER CLAMP CONTROL



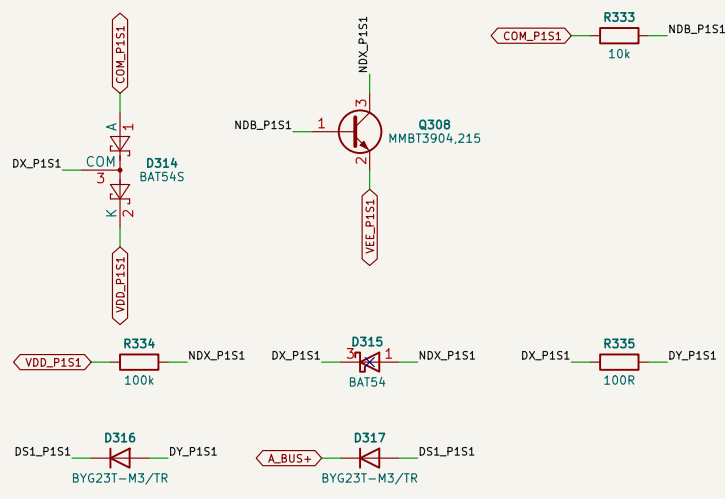
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S1: SHORT-CIRCUIT BOOSTER



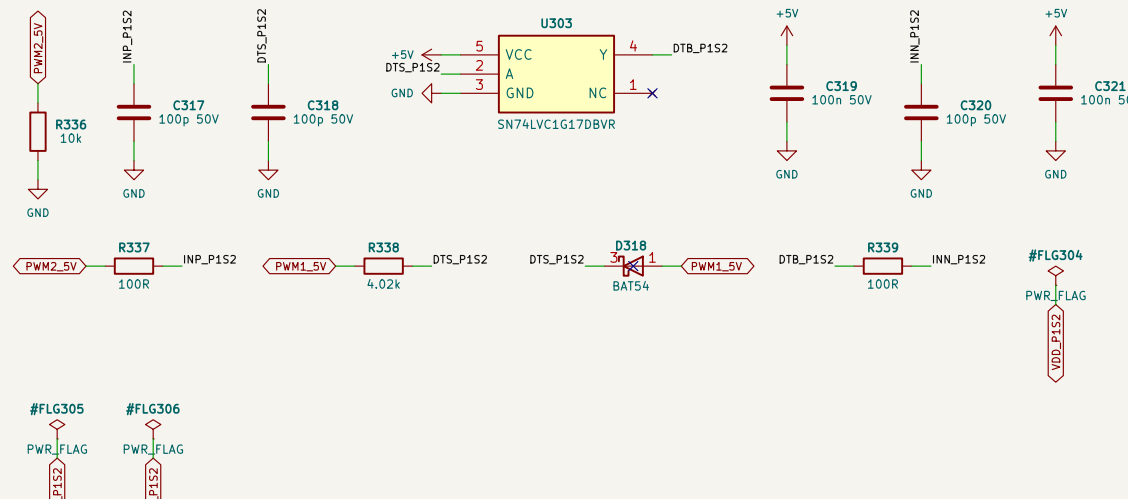
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device A03A00A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P1S1: DESAT



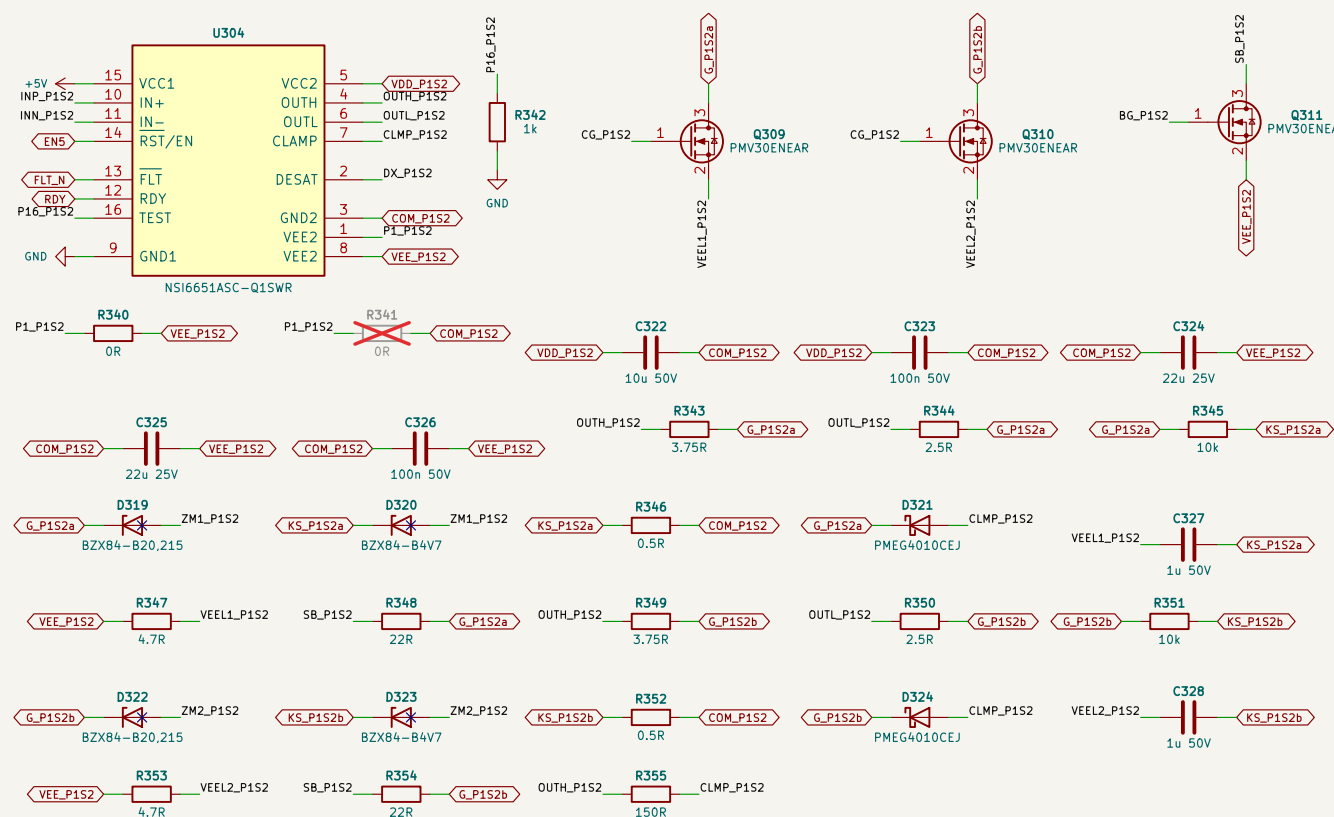
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Calk footprint DMP.

P1S2: LOGIC SIDE



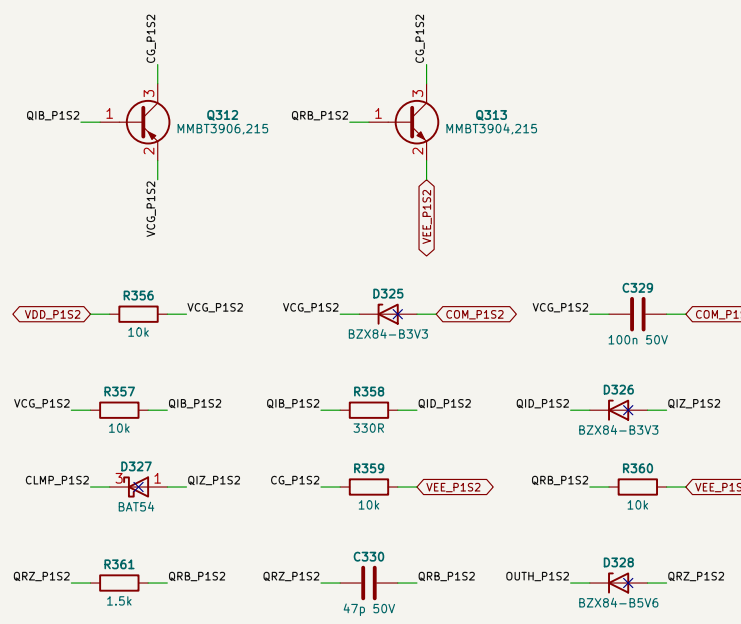
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S2: DRIVER AND 2 GATES



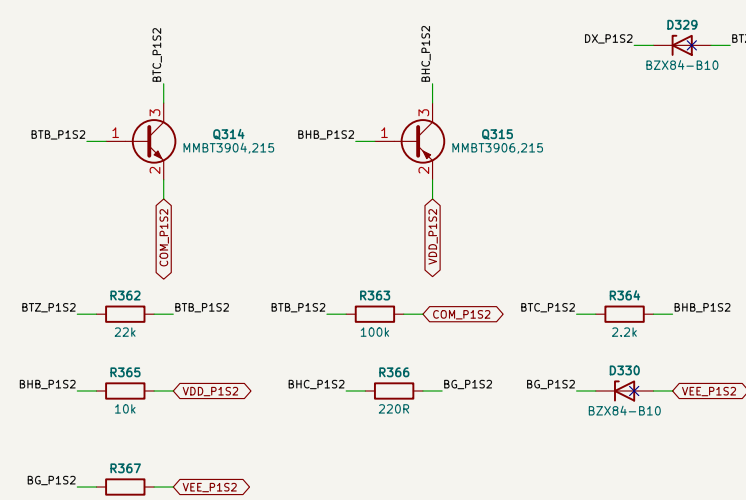
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S2: MILLER CLAMP CONTROL



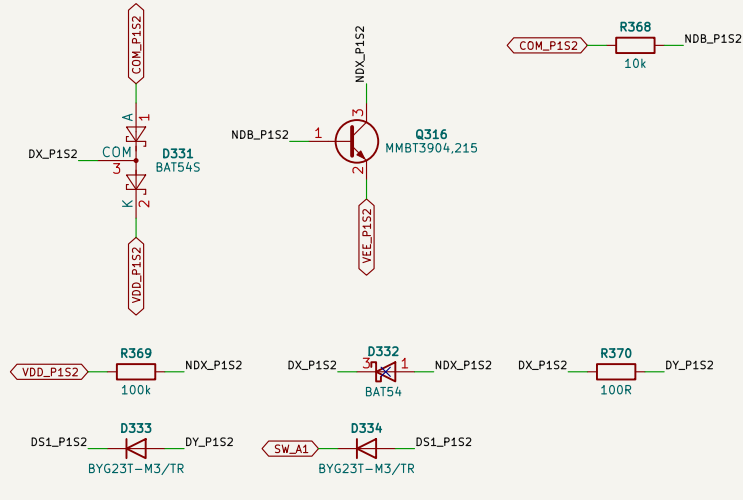
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device A03A00A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P1S2: DESAT



DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Calk footprint DMP.

LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10%) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per slider; board coated to P01 (IC-22). Miller hold (ngapic, groy): die +1.41 V vs VGS(n) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 4

Schematics authored by Chinmoy Bhuyhan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /03 Phase 1 leg A gate drive (S1, S2)/
File: 03_ph1_gda.kicad_sch

Title: PV-PWR power board - Phase 1 leg A gate drive (S1, S2)

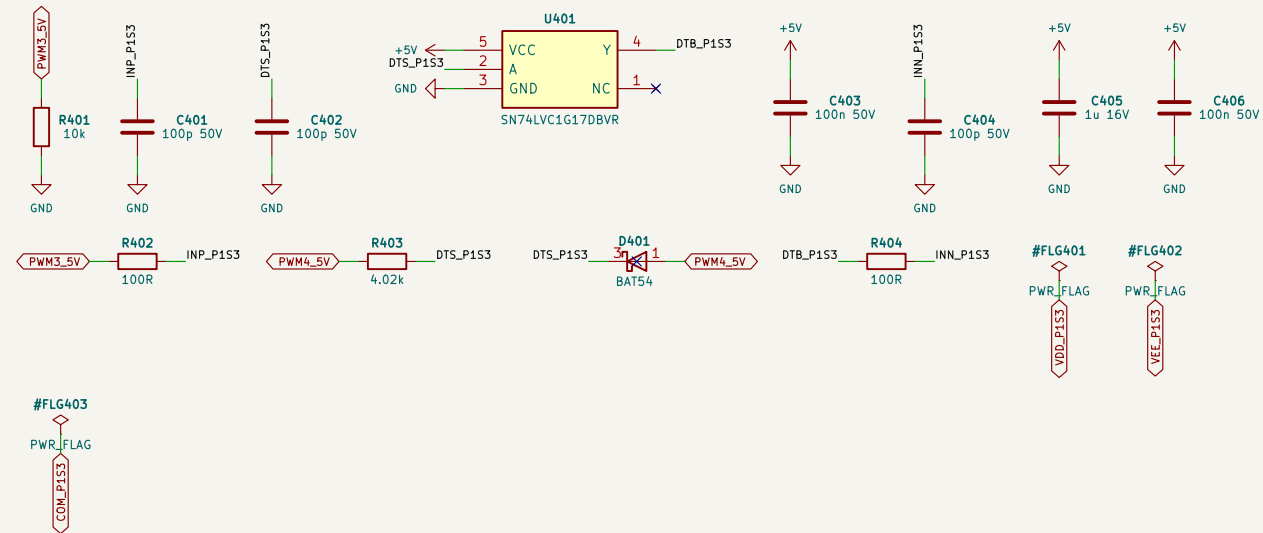
Size: A2 Date: 2026-10-05

KiCad E.D.A. 10.0.6

Rev: A2

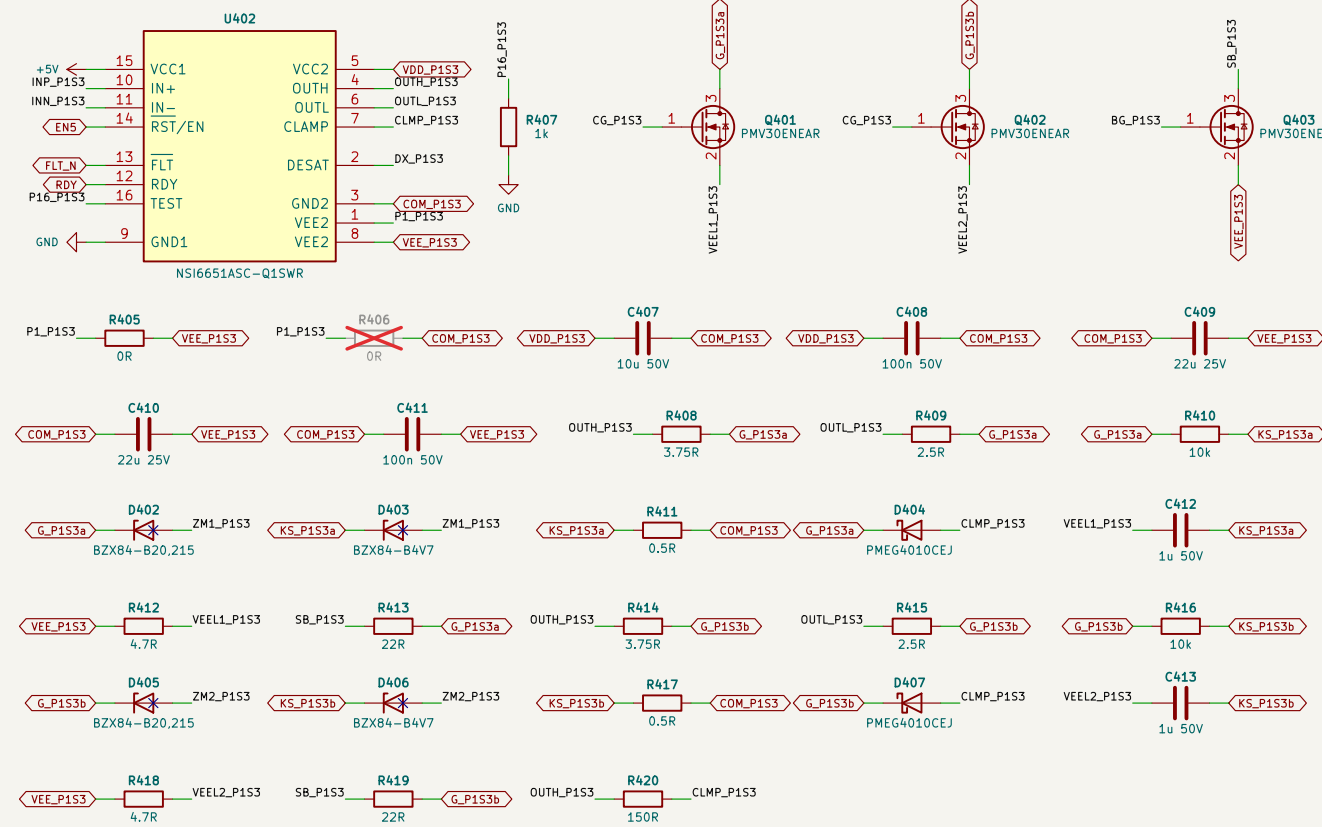
Id: 4/23

P1S3: LOGIC SIDE



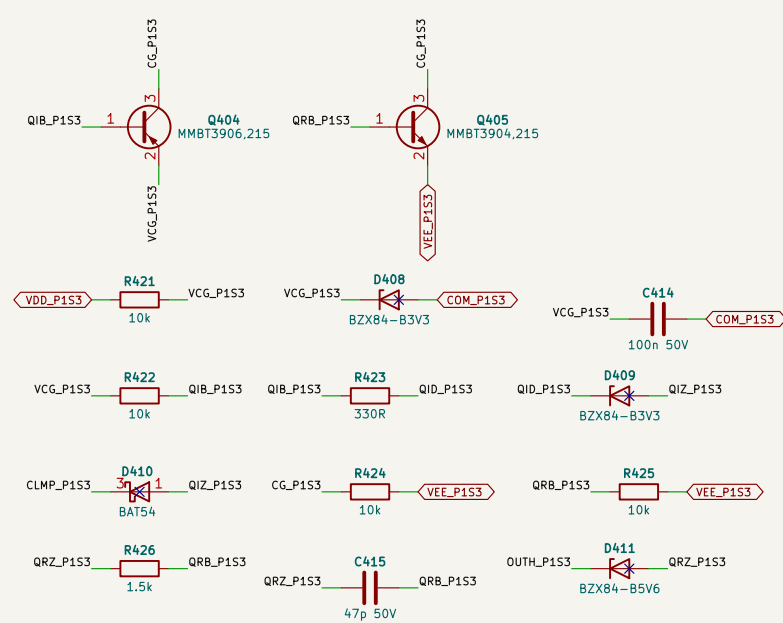
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



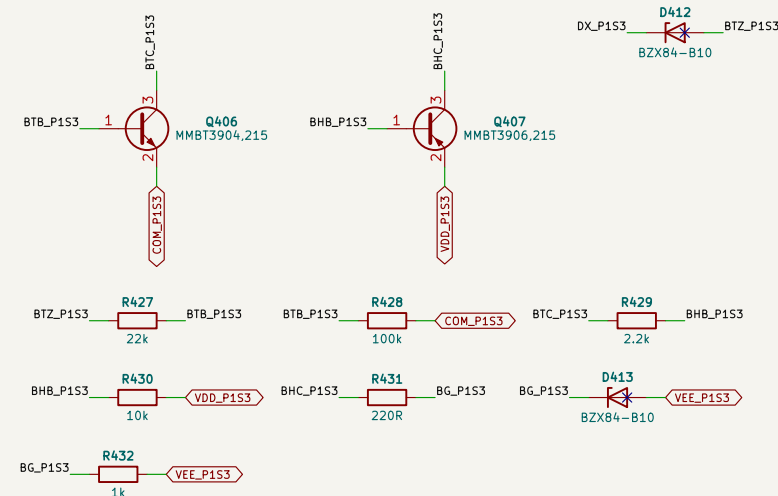
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



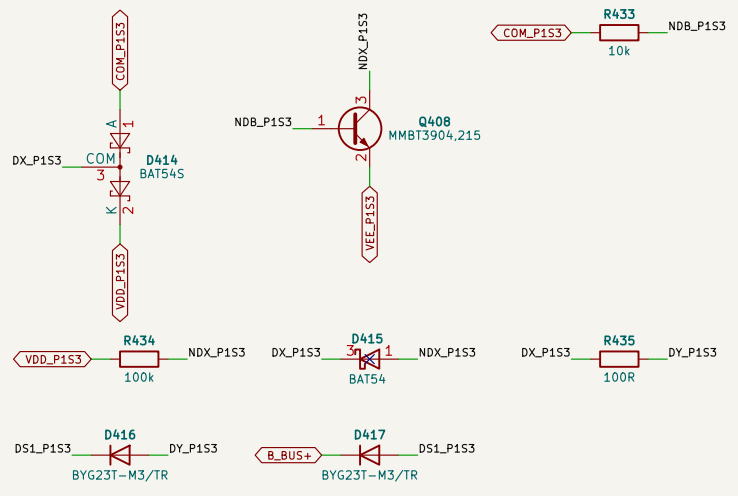
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER



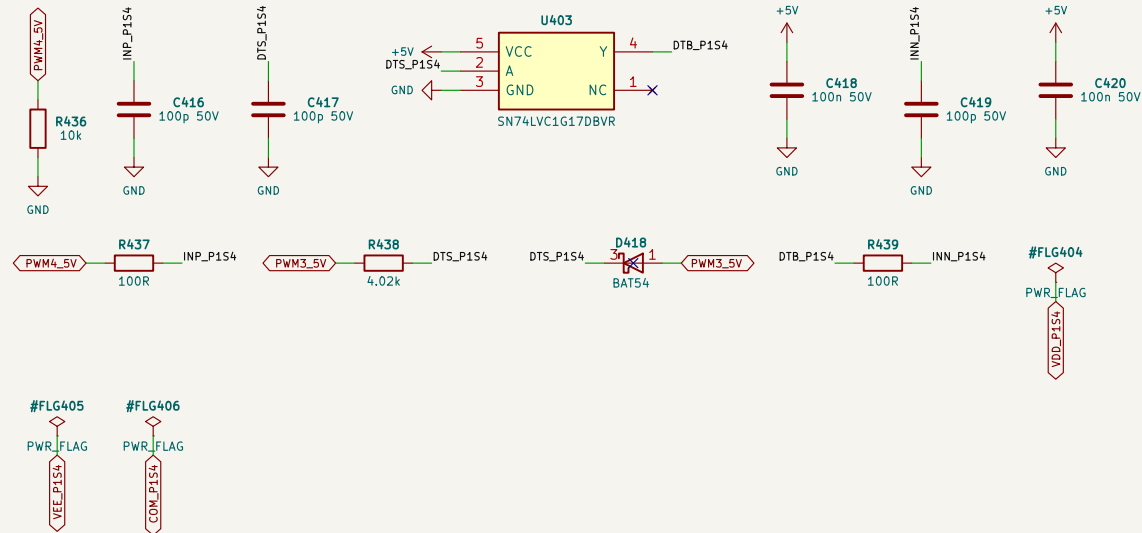
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P1S3: DESAT



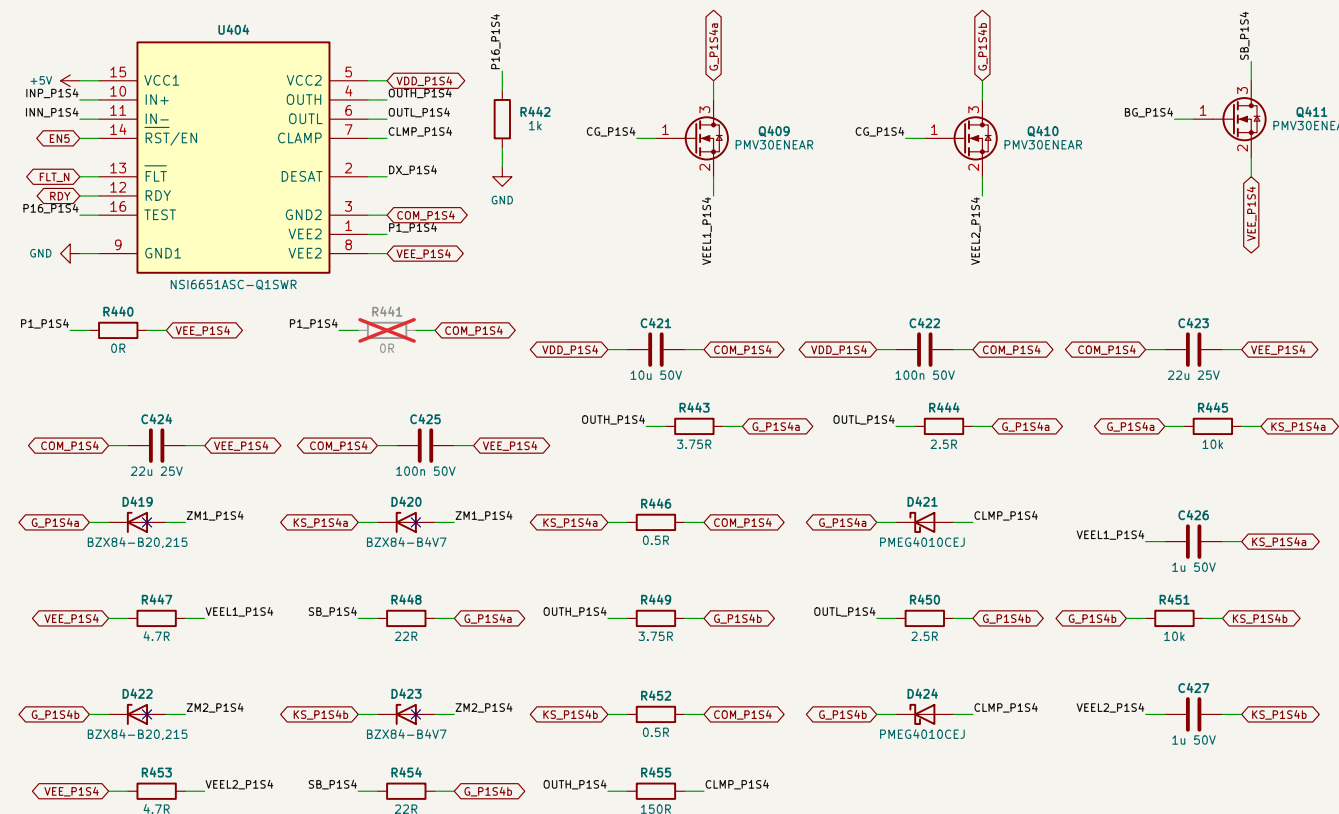
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P1S4: LOGIC SIDE



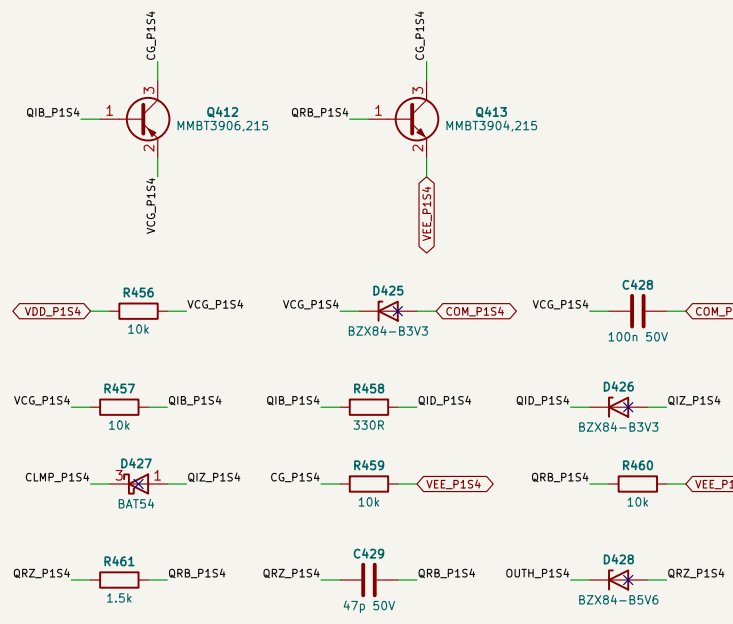
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S4: DRIVER AND 2 GATES



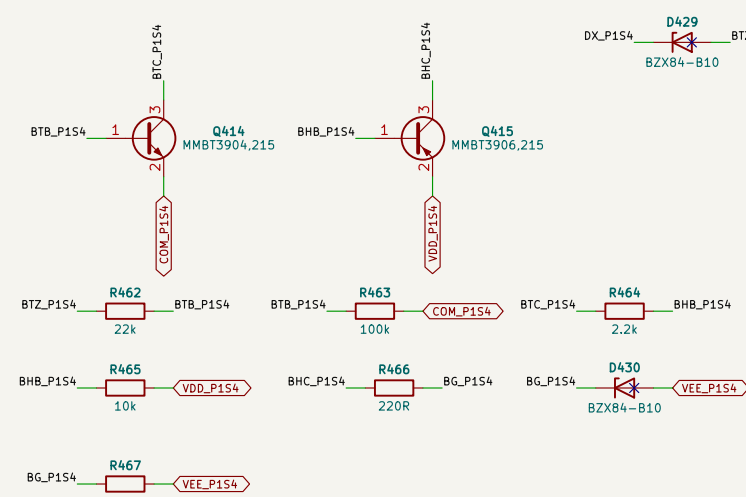
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S4: MILLER CLAMP CONTROL



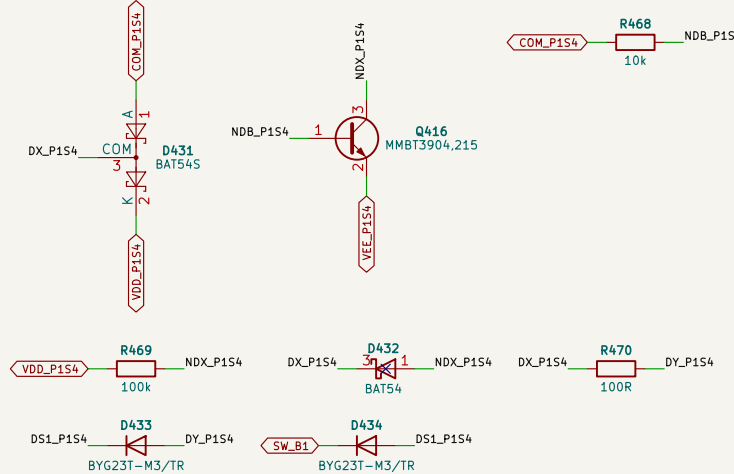
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below ~ -1.2 V
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P1S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprints DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length ($\pm 10\%$) from the COM star; Ron, Roff, Kelvin R, 10k. Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop ≤ 1 nH. DESAT string ≥ 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die ± 1.41 V vs VGS(th) min 1.94 V at 175 C – BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 5

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE base

Sheet: /04 Phase 1 leg B gate drive (S3, S4)/

File: 04_ph1_gdb.kicad_sch

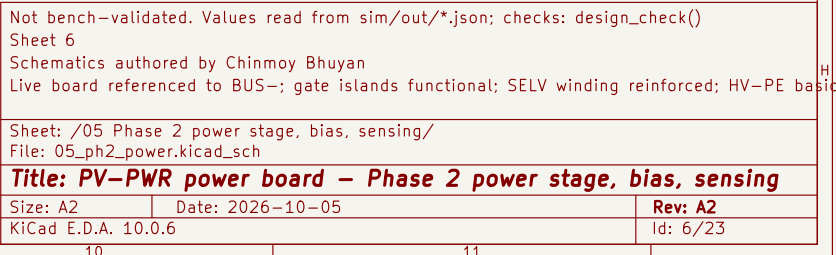
Title: PV-PWR power board – Phase 1 leg B gate drive (S3, S4)

Size: A2

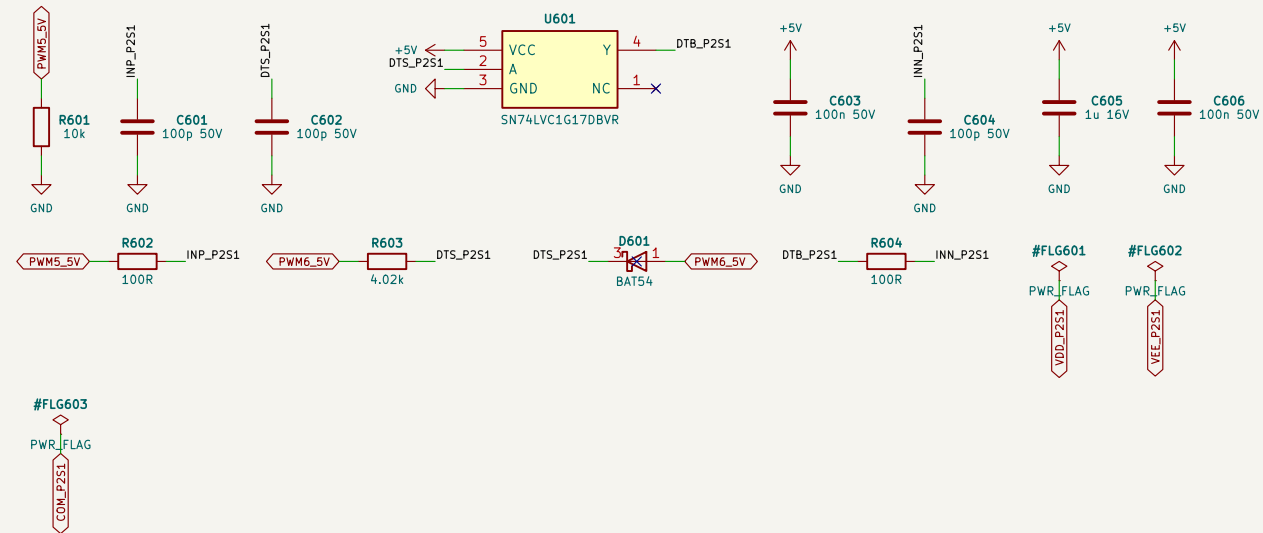
Date: 2026-10-05

Rev: A2

Id: 5423

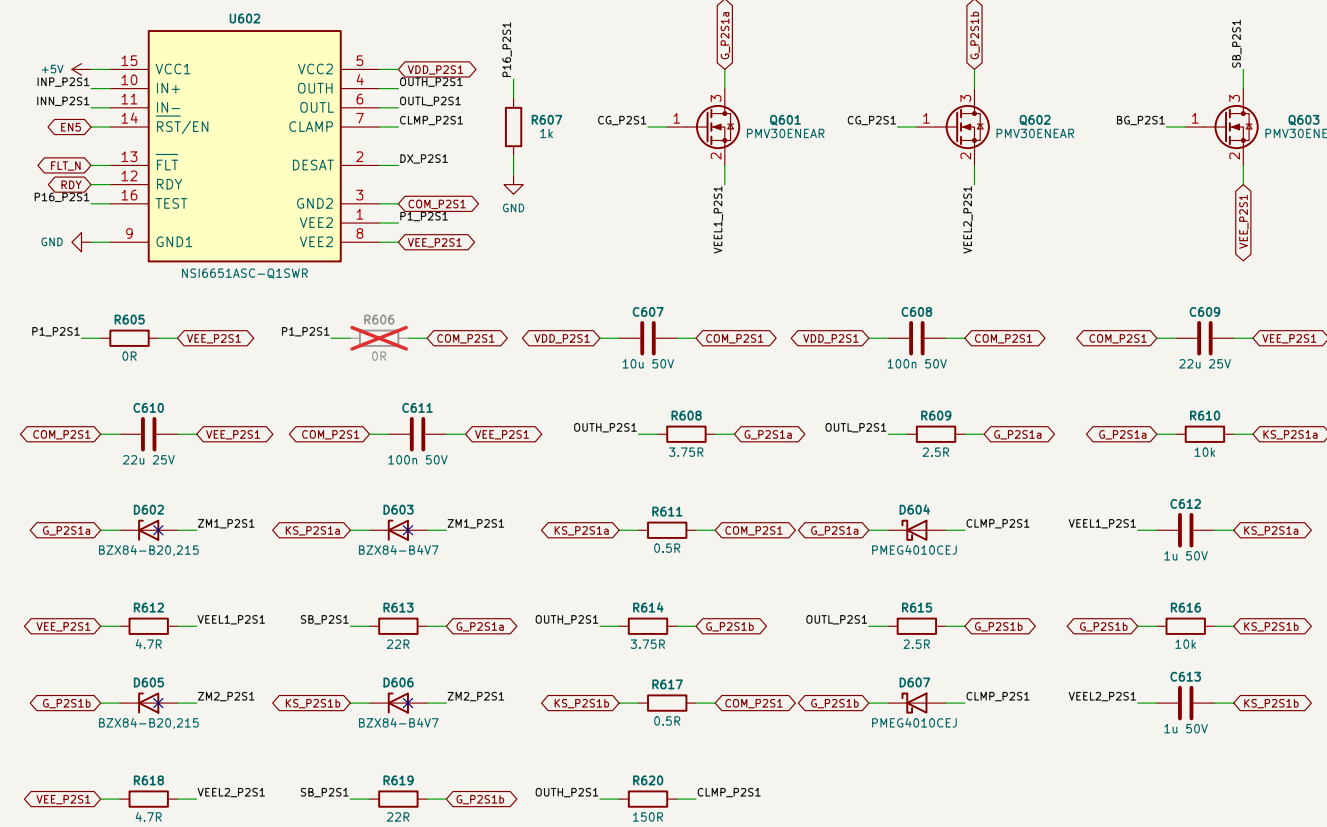


P2S1: LOGIC SIDE



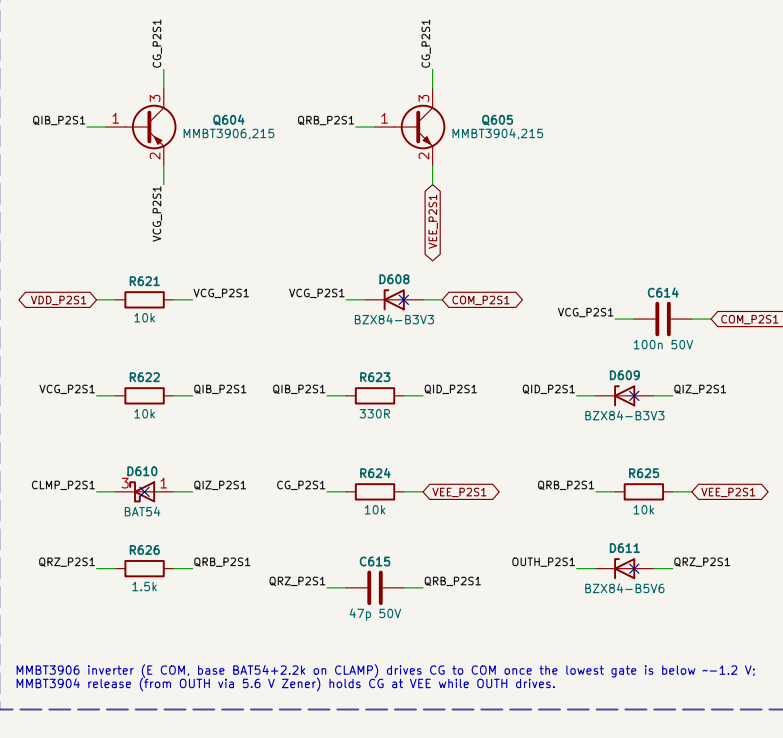
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



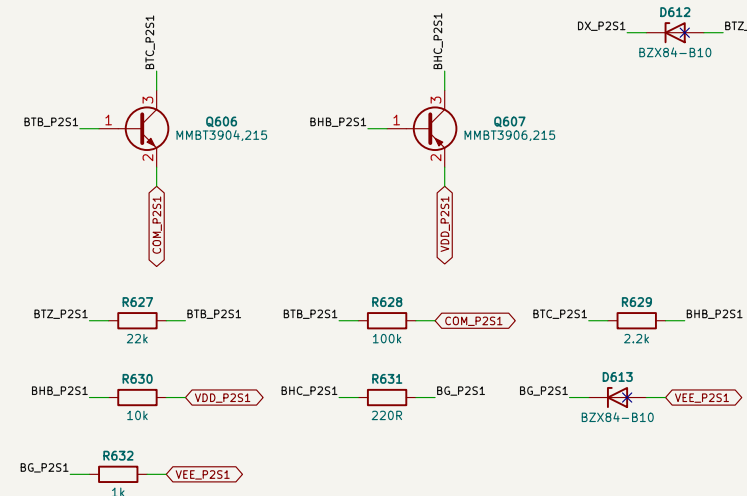
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL



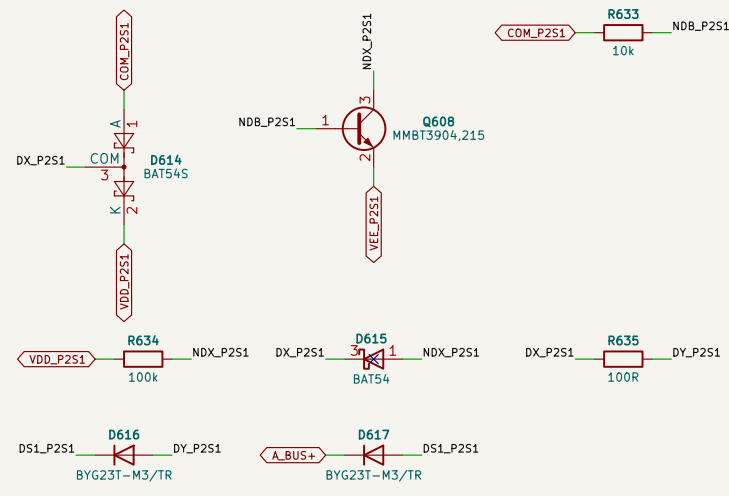
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V.
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER



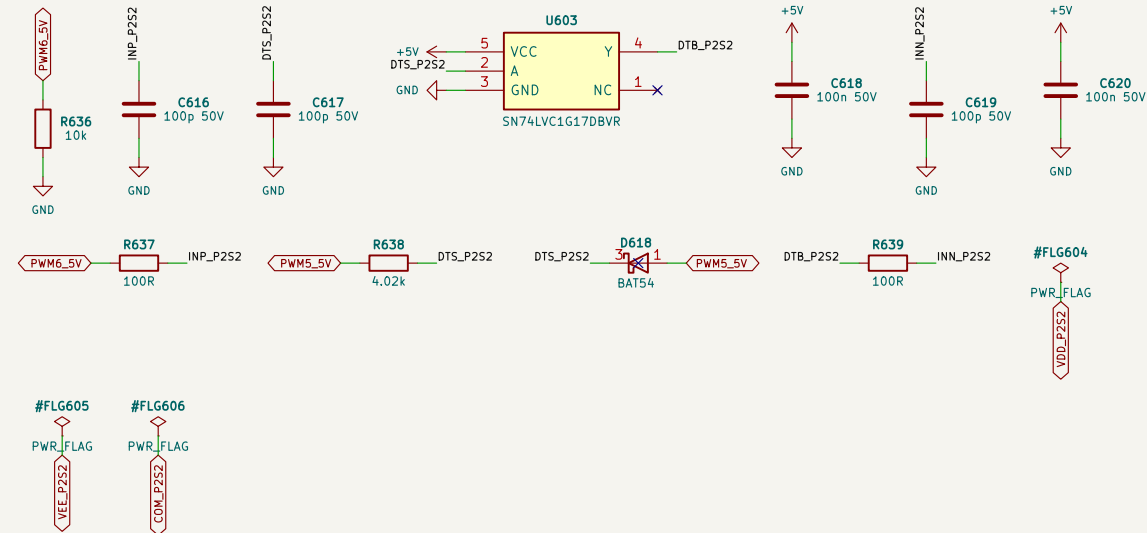
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P2S1: DESAT



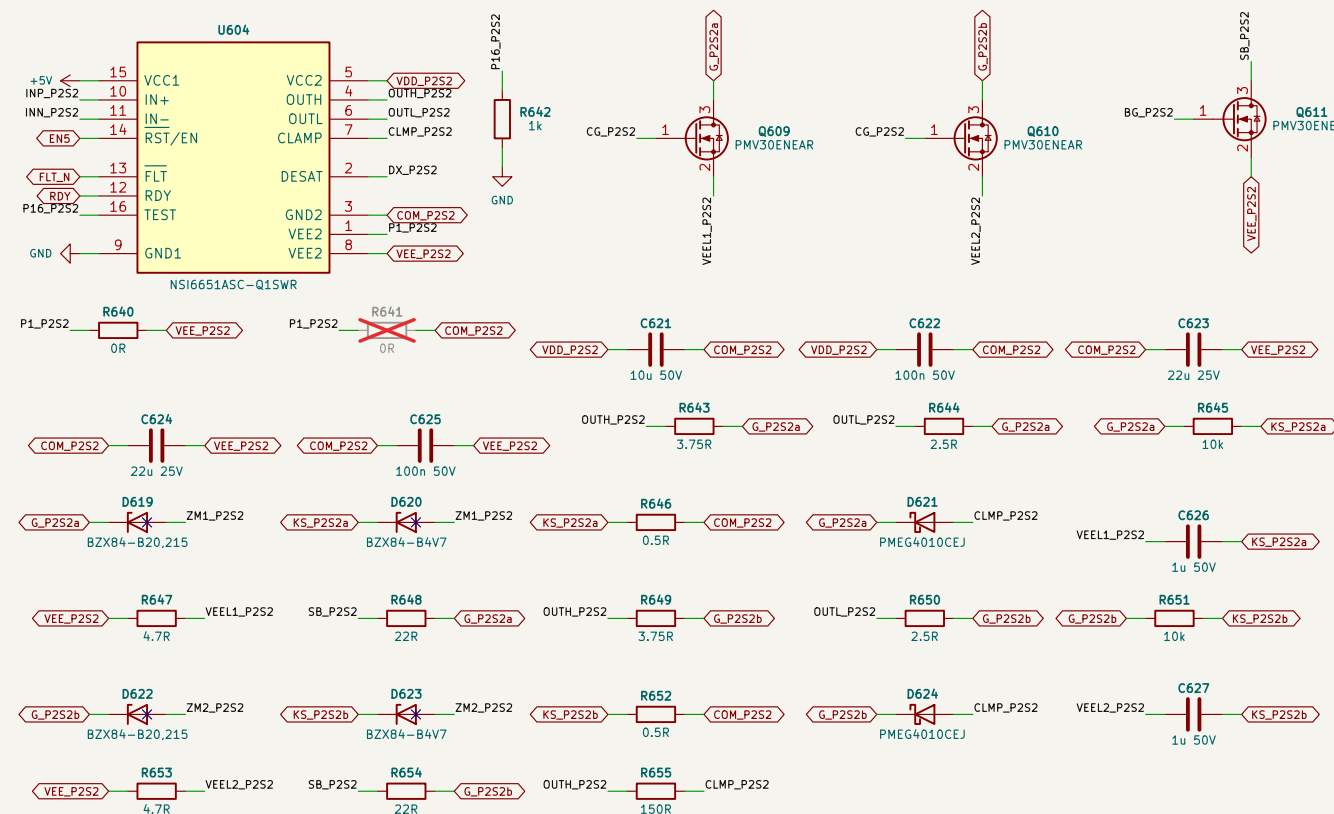
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P2S2: LOGIC SIDE



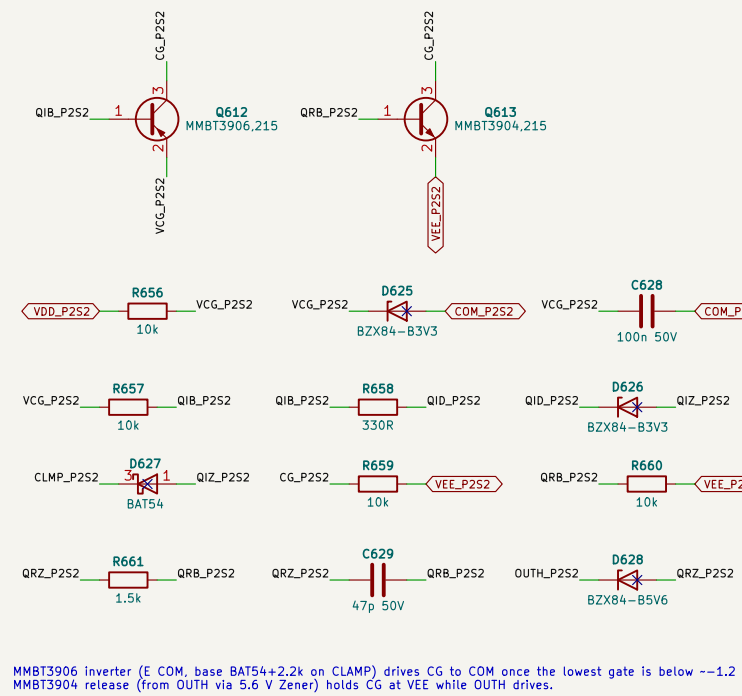
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S2: DRIVER AND 2 GATES



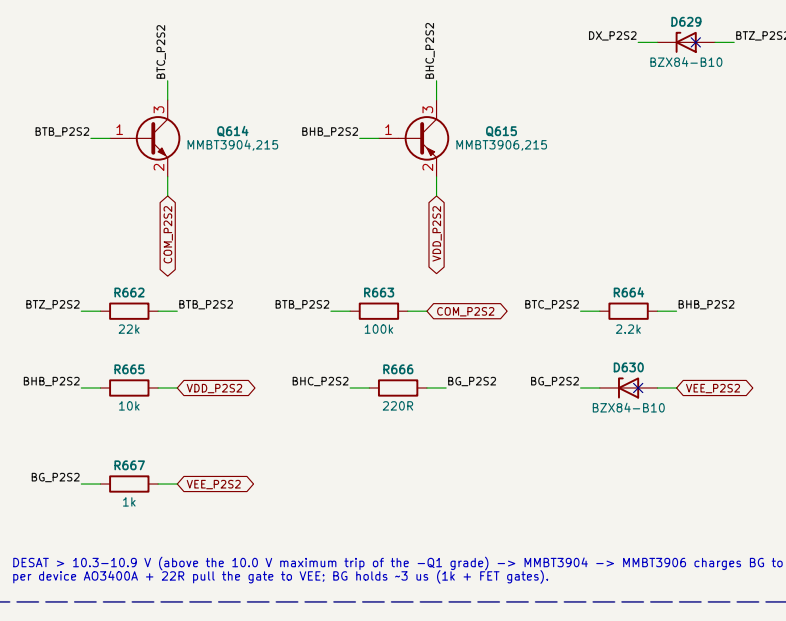
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S2: MILLER CLAMP CONTROL



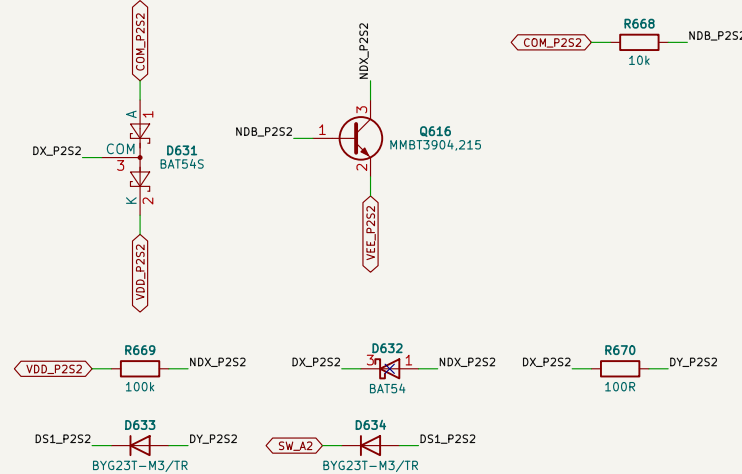
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P2S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length ($\pm 10\%$) from the COM star; Ron, Roff, Kelvin R, 10k. Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop ≤ 1 nH. DESAT string ≥ 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die ± 1.41 V vs VGS(th) min 1.94 V at 175 C – BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 7

Schematics authored by Chinmoy Bhuyan

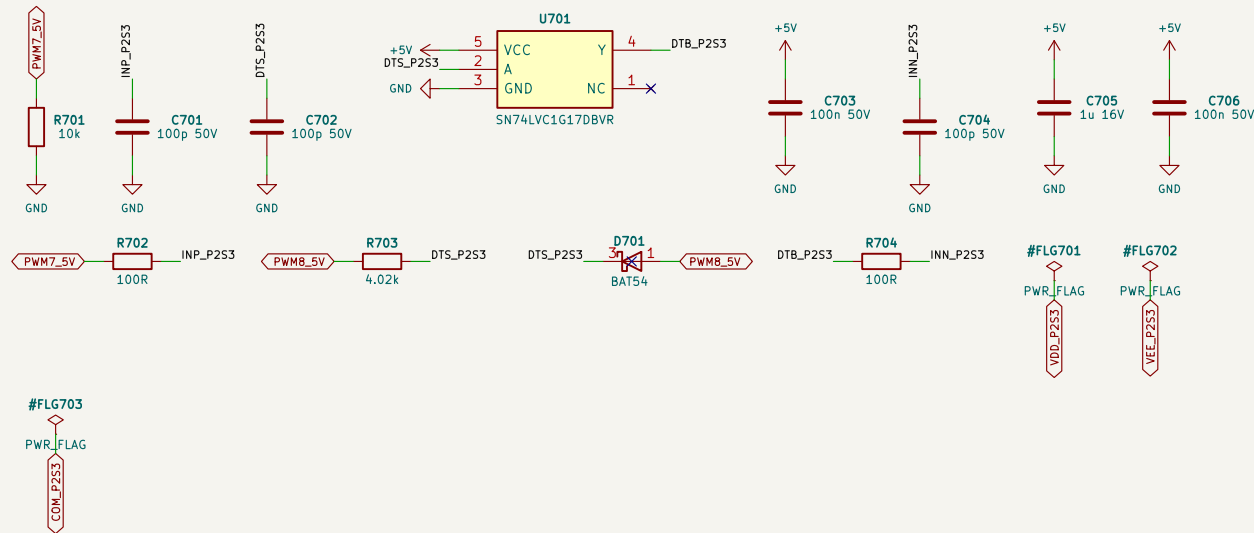
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE base

Sheet: /06 Phase 2 leg A gate drive (S1, S2)/

Title: PV-PWR power board – Phase 2 leg A gate drive (S1, S2)

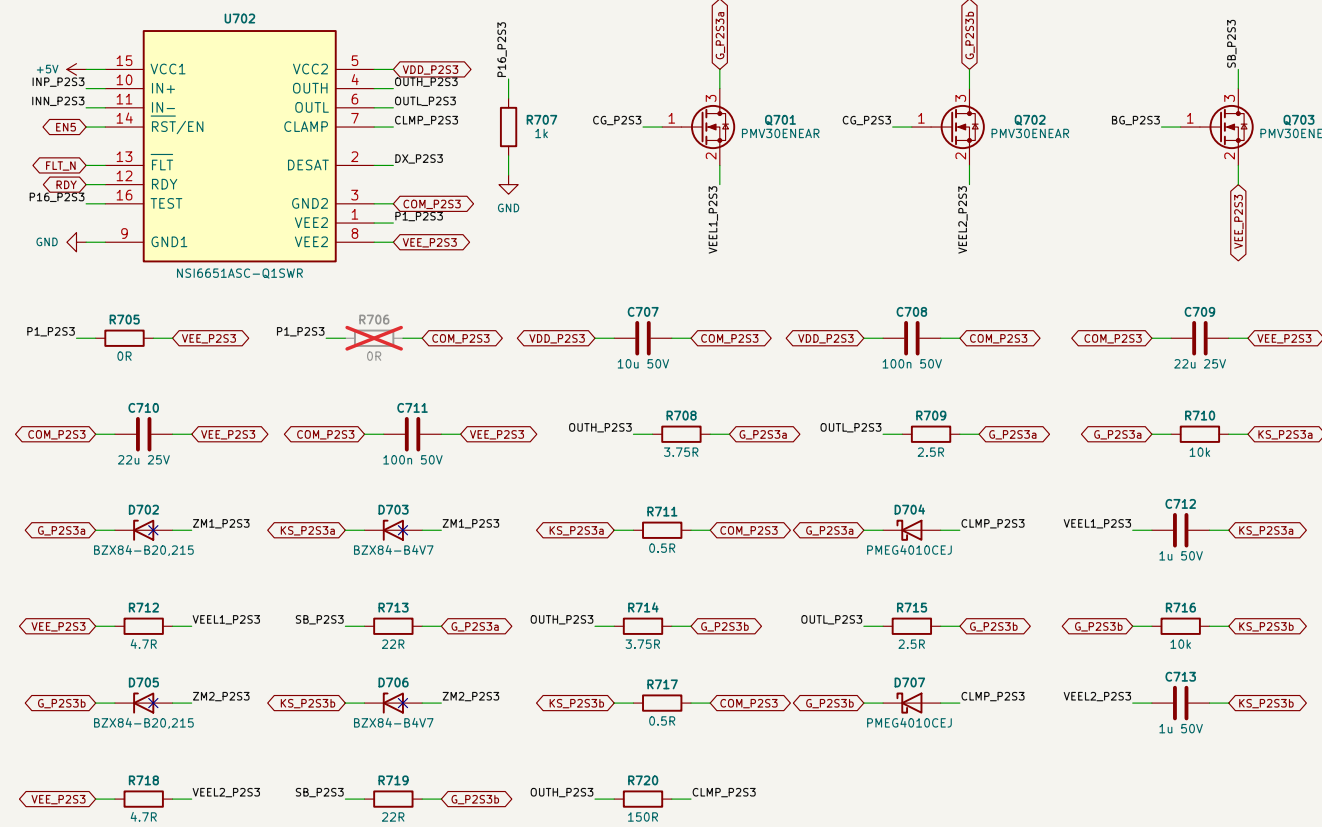
Size: A2	Date: 2026-10-05	Rev: A2
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P2S3: LOGIC SIDE



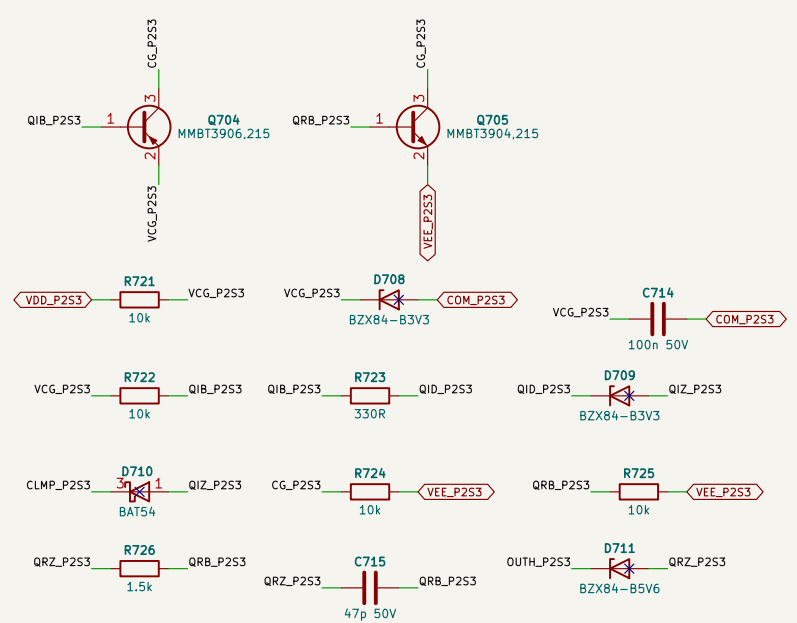
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



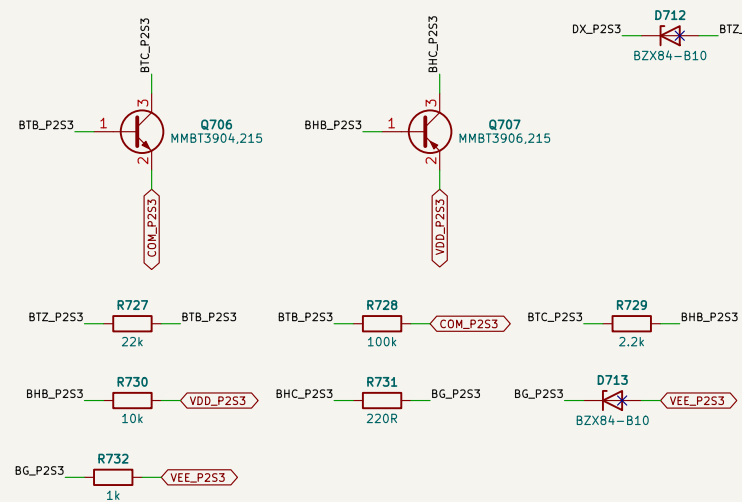
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL



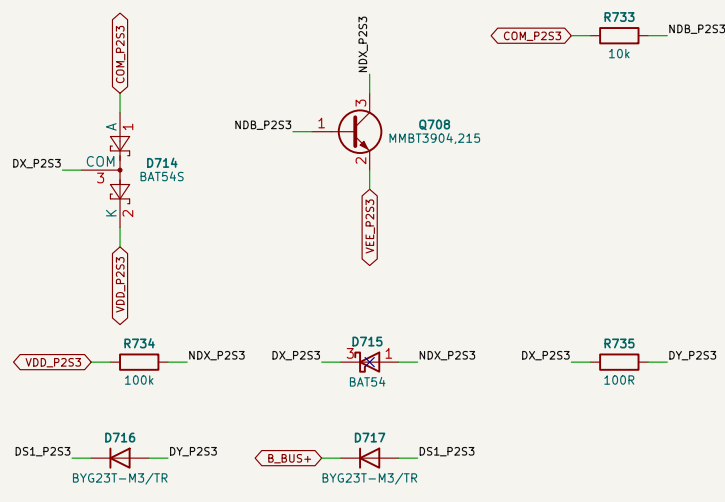
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V:
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S3: SHORT-CIRCUIT BOOSTER



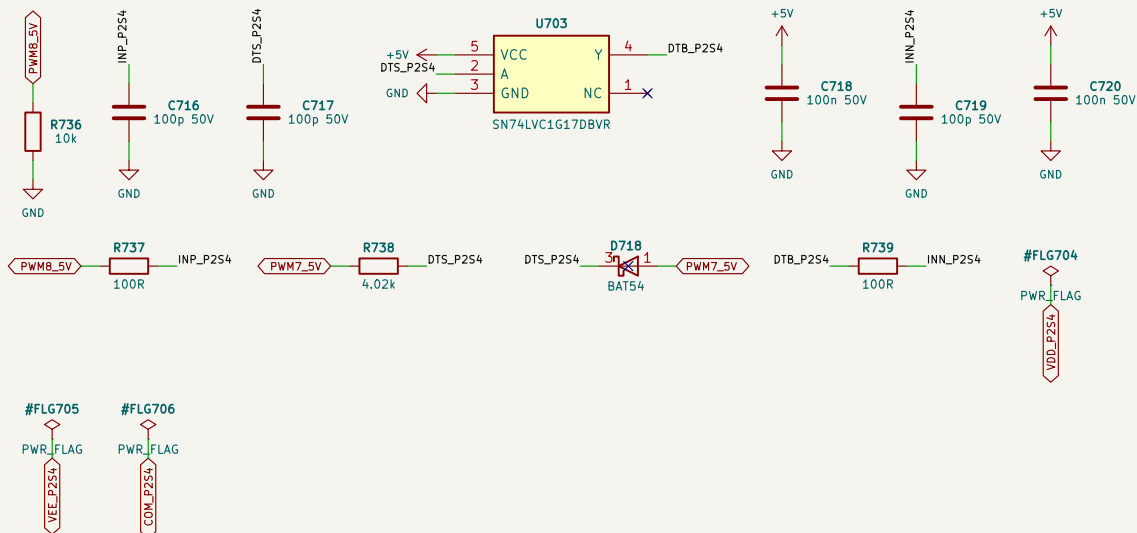
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V:
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P2S3: DESAT



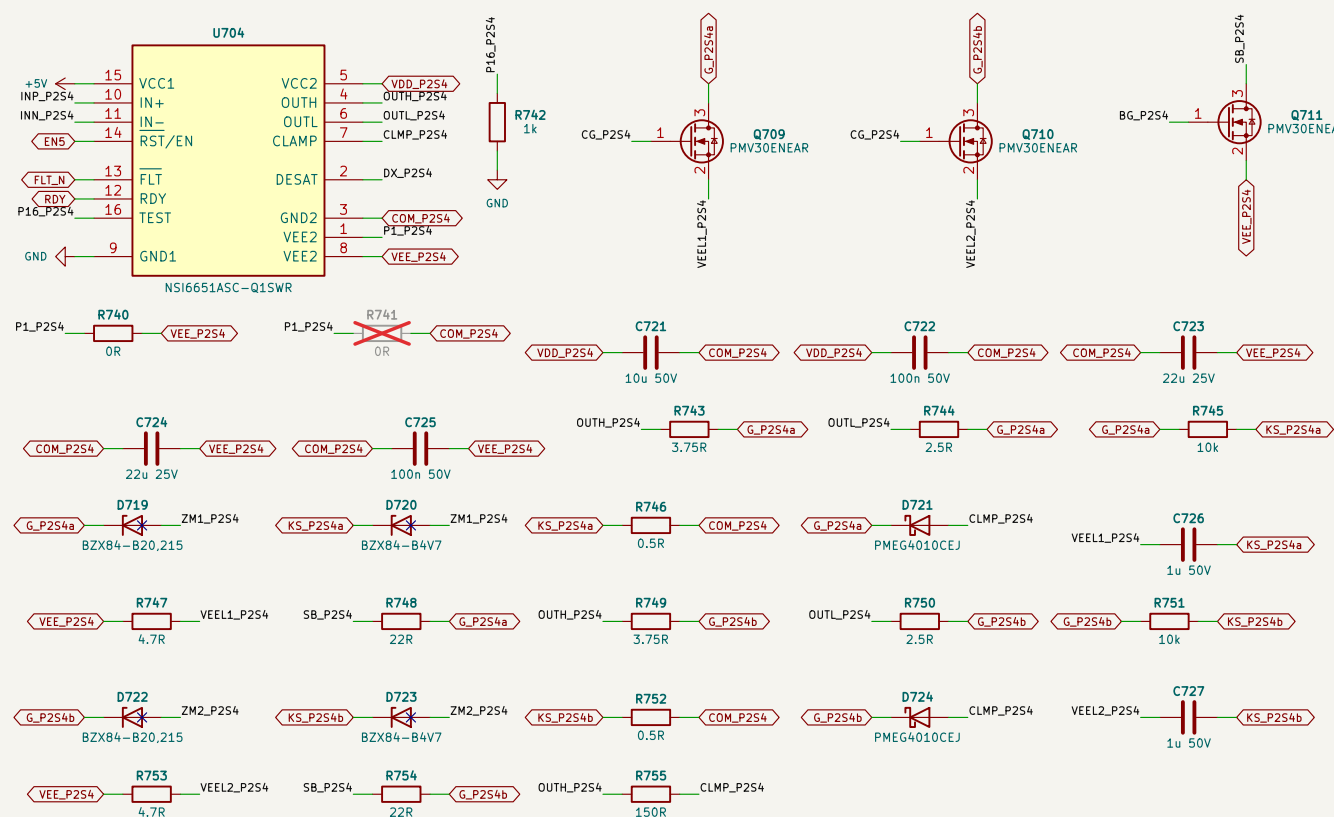
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Calk footprint DMP.

P2S4: LOGIC SIDE



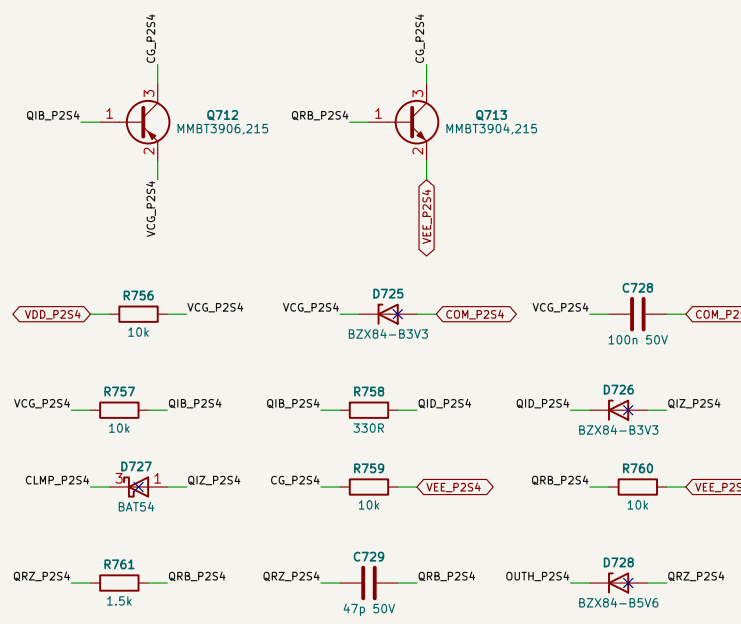
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S4: DRIVER AND 2 GATES



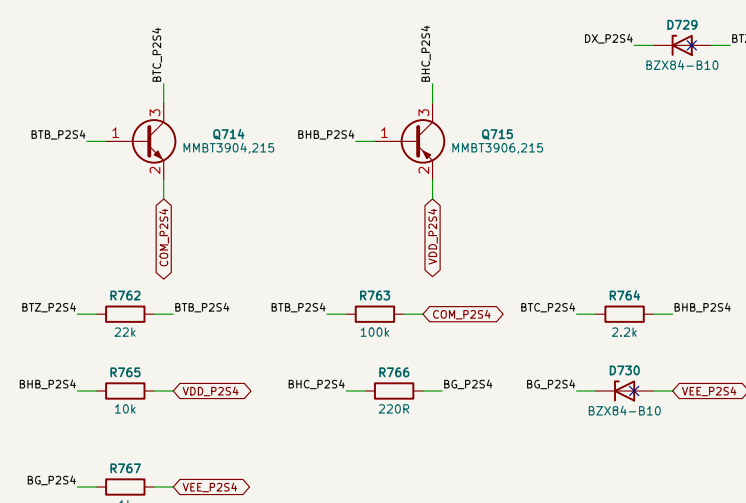
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S4: MILLER CLAMP CONTROL



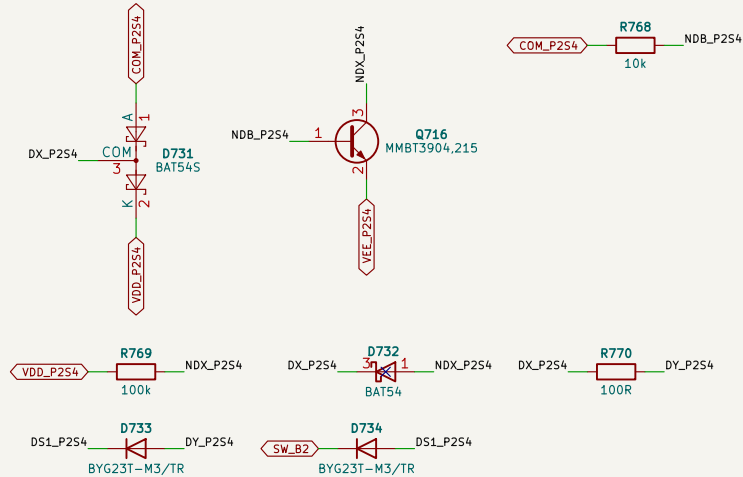
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V:
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V:
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P2S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Calk footprint DMP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per slider; board coated to P01 (IC-22). Miller hold (nagpipe, gero): die +1.4 V vs VGS(n) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 8

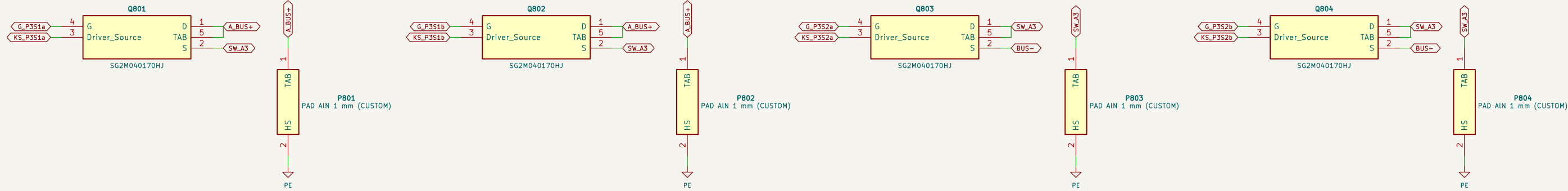
Schematics authored by Chinnmoy Bhuyhan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /07 Phase 2 leg B gate drive (S3, S4)/
File: 07_ph2-gdb.kicad_sch

Title: PV-PWR power board - Phase 2 leg B gate drive (S3, S4)

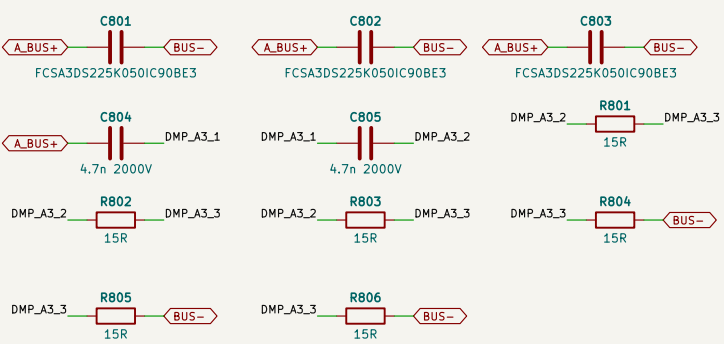
Size: A2 Date: 2026-10-05 Rev: A2
KiCad E.D.A. 10.0.6 Id: 8/23

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



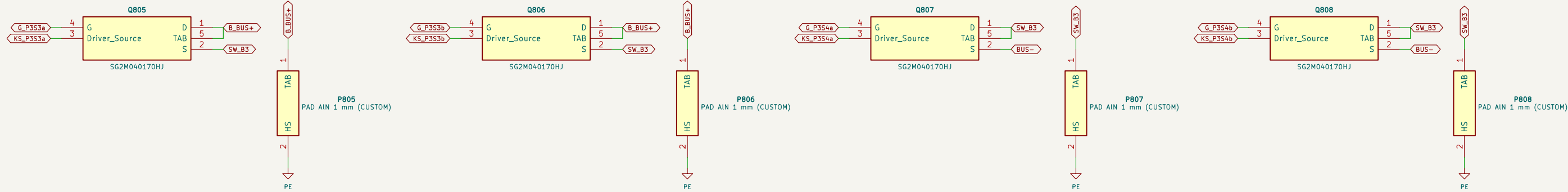
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



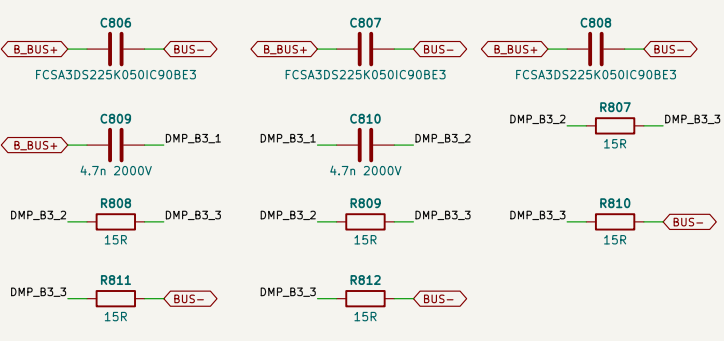
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



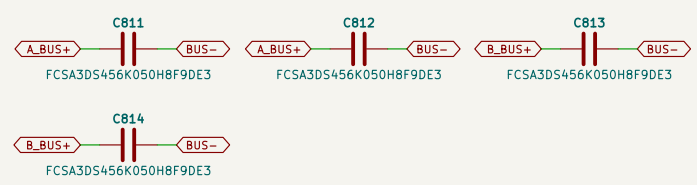
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



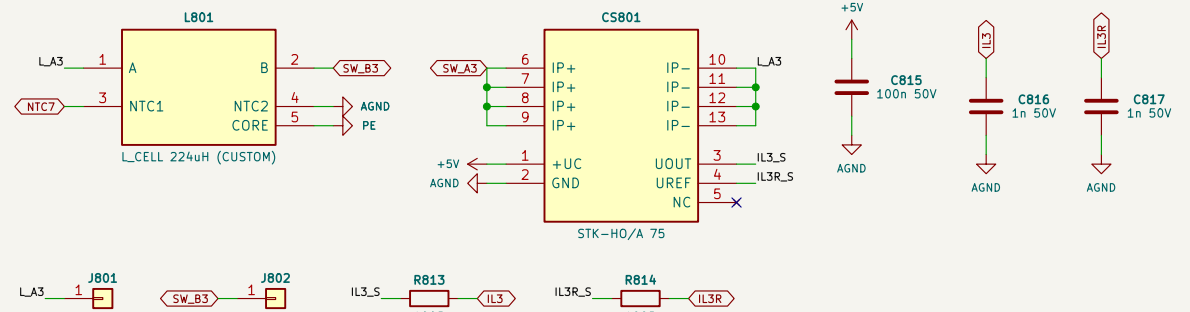
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



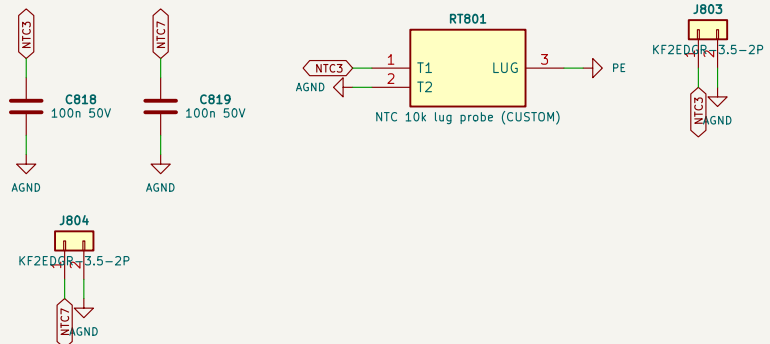
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



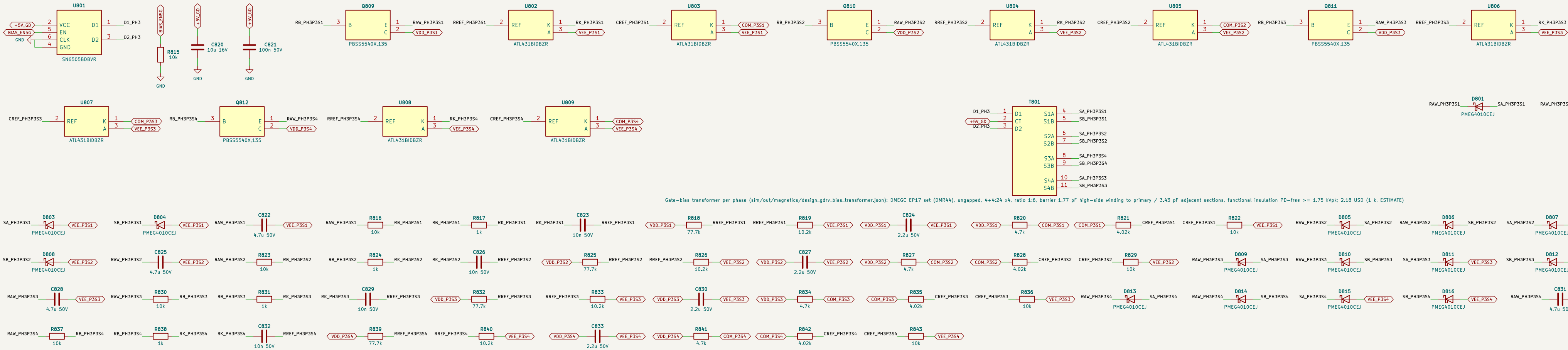
Lead A from SW_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out + = SW_A -> SW_B) to stud LA3: lead B on stud SW_B3. IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A: 100R + 1 nF at the source. IL3R = the sensor's own Uref (pin 4). 100R + 1 nF: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



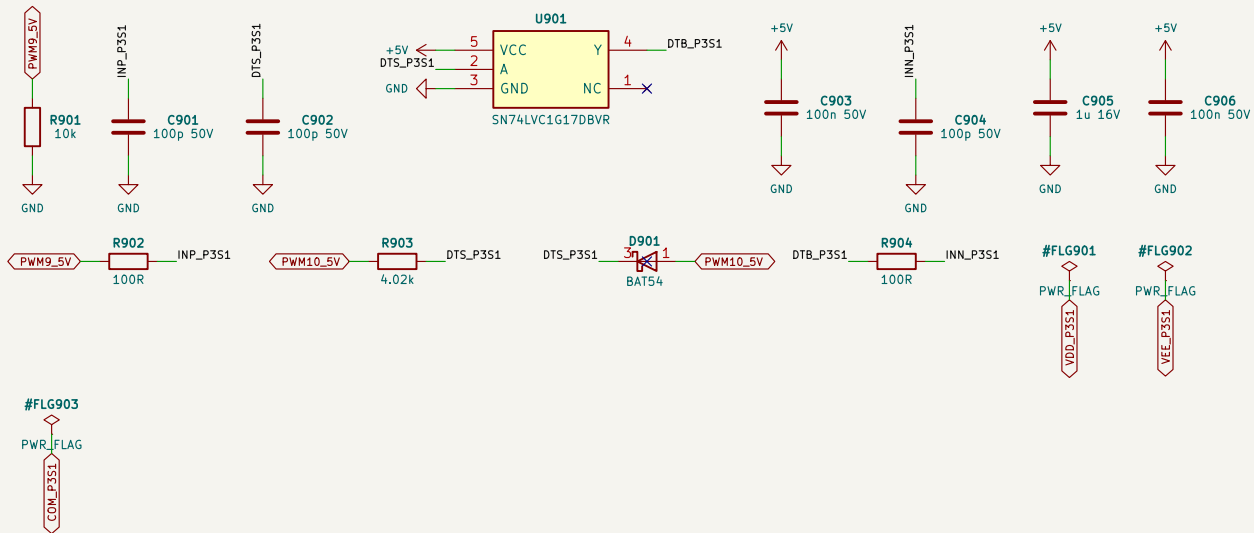
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10). NTC7 = inductor winding NTC (functional to the winding). 100 nF at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



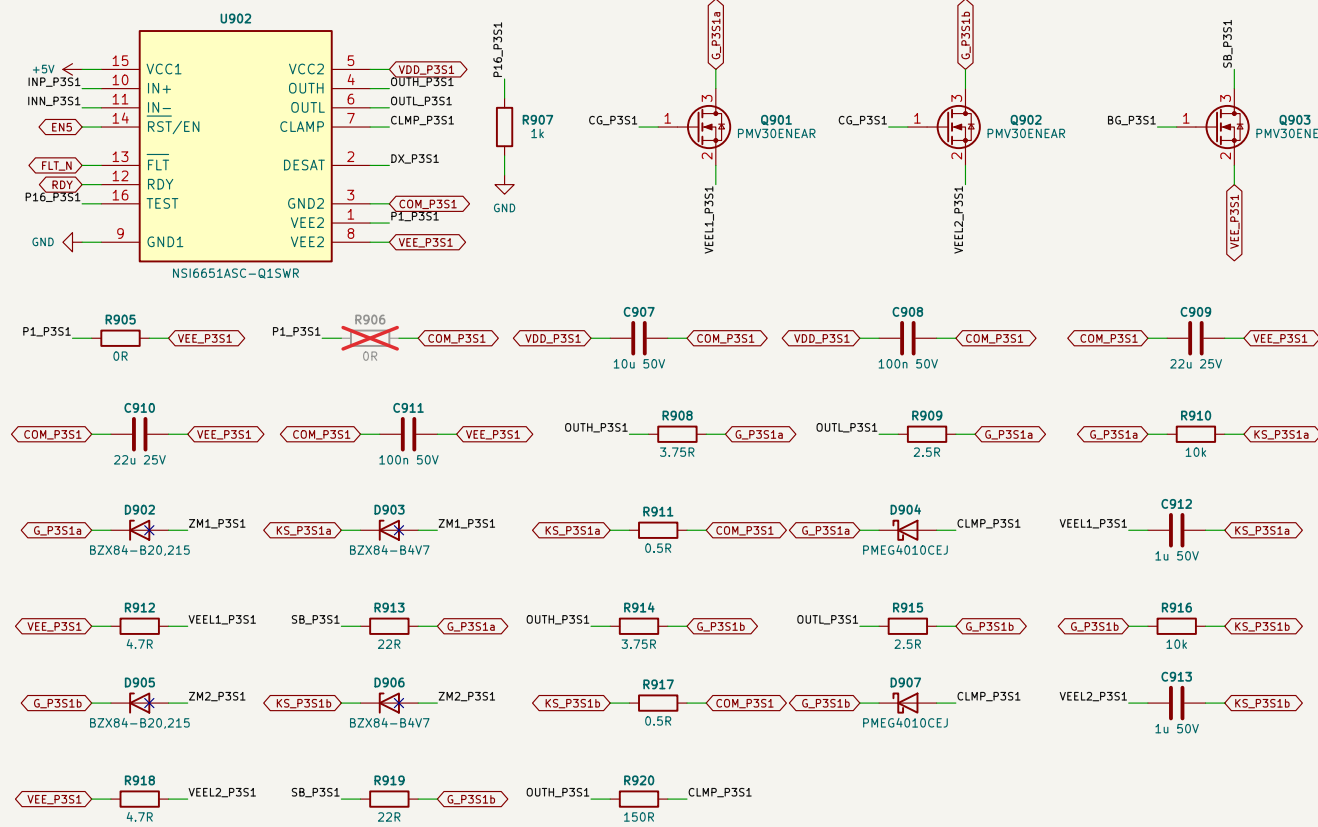
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T.BIAS4 custom, functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



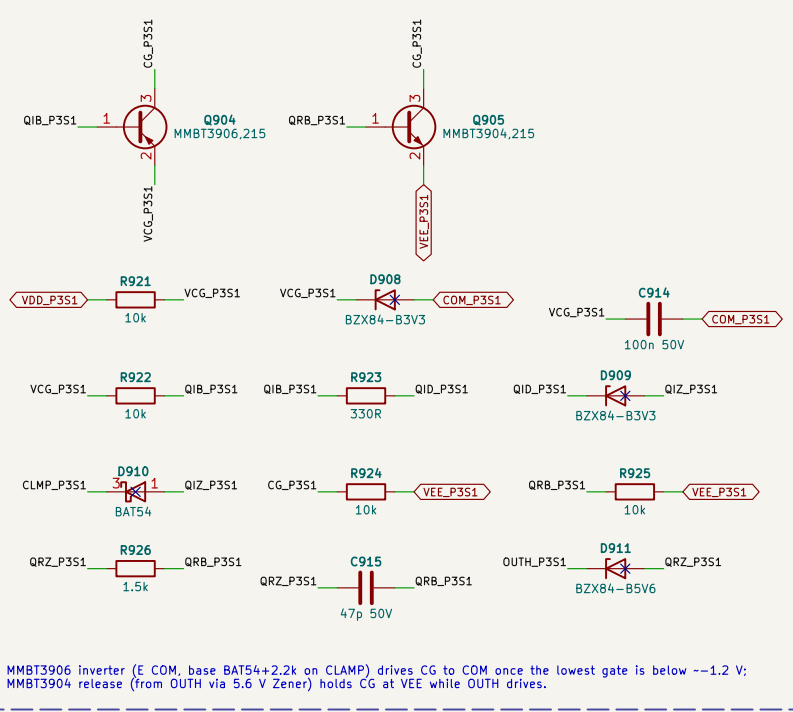
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



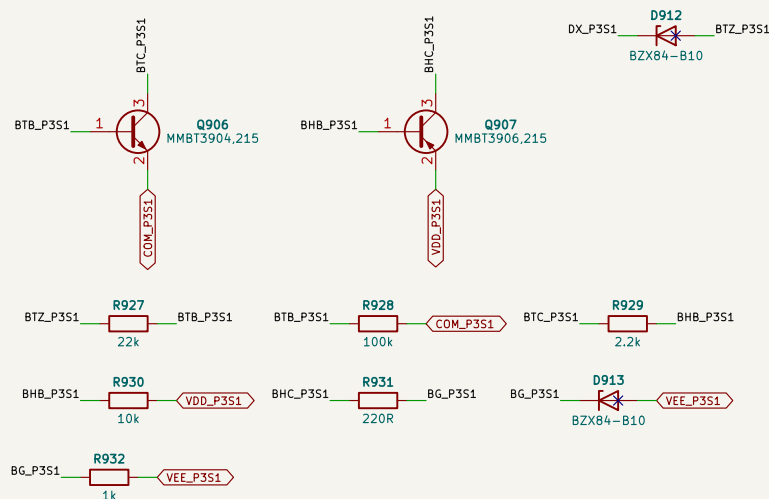
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



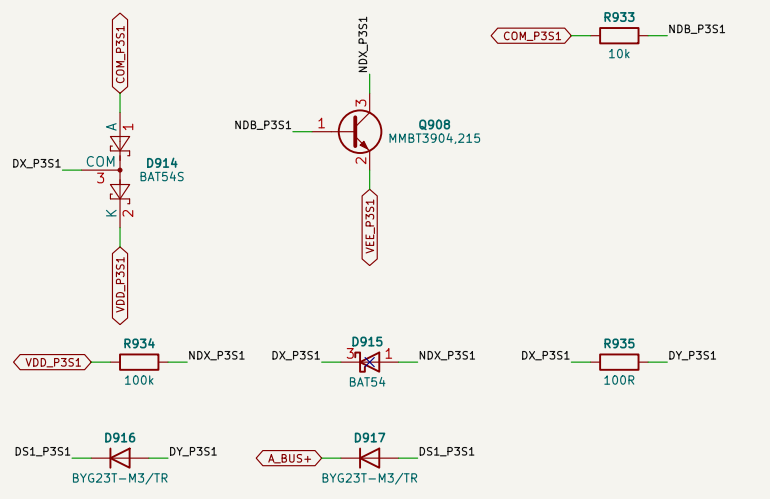
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



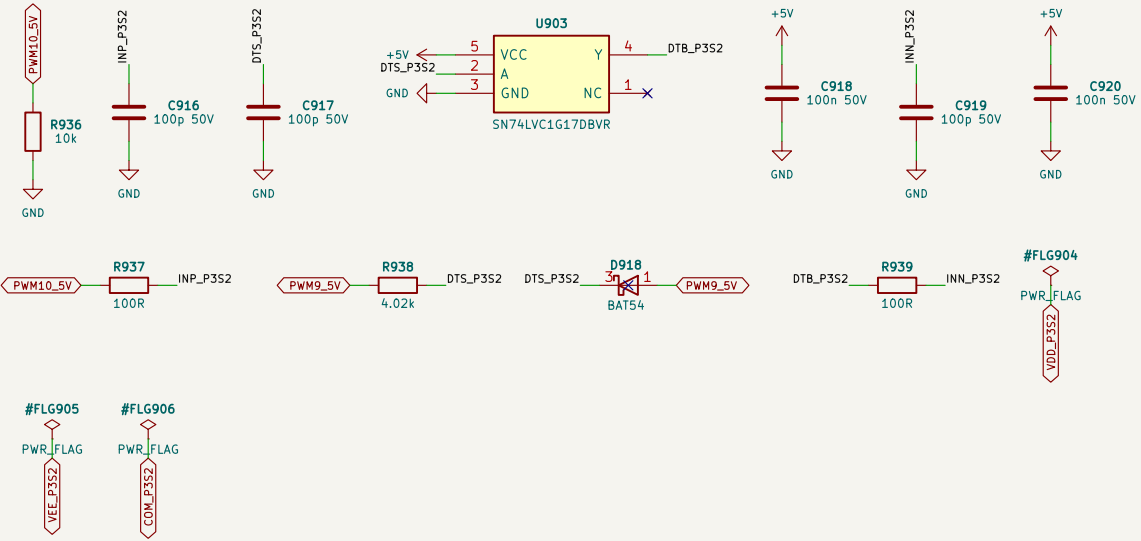
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds -3 us (A + FET gates).

P3S1: DESAT



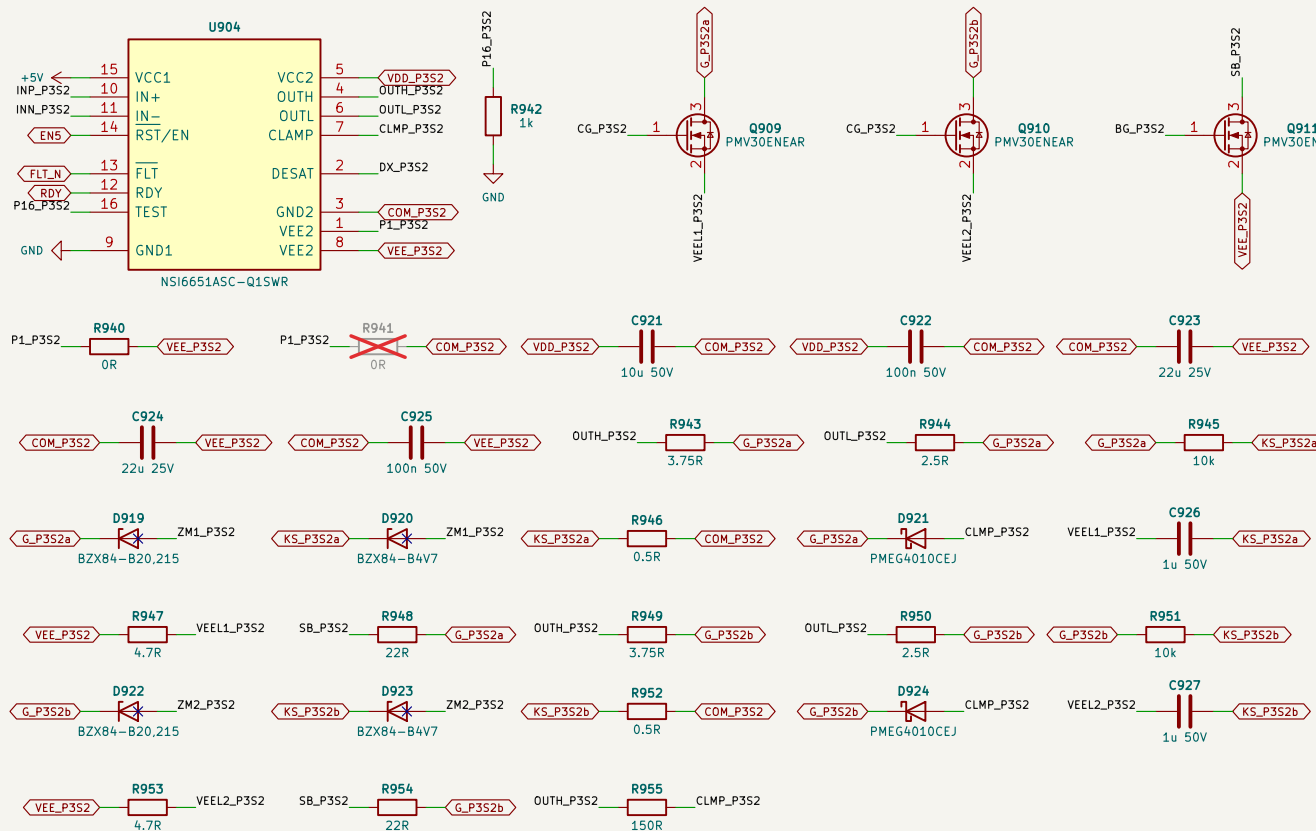
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P3S2: LOGIC SIDE



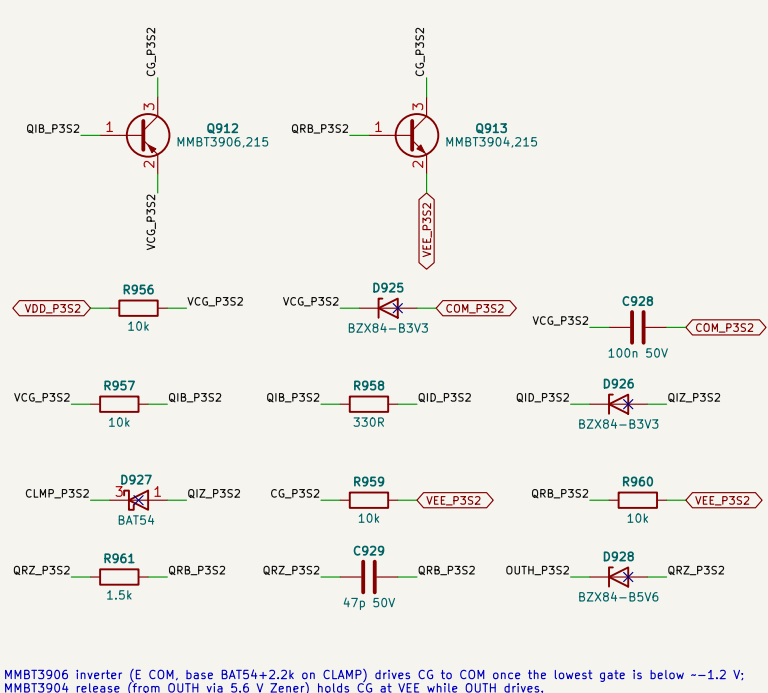
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



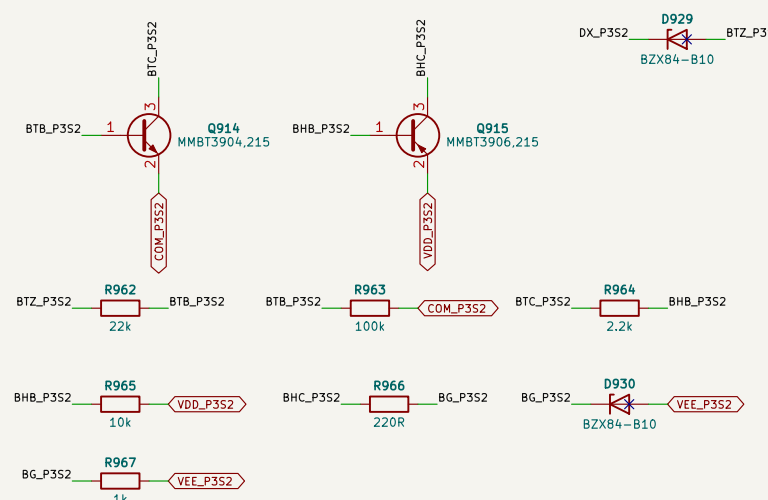
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



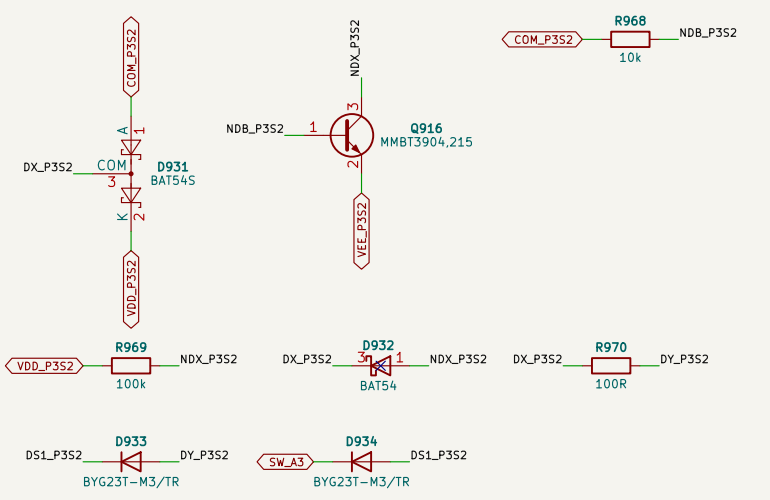
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S2: DESAT

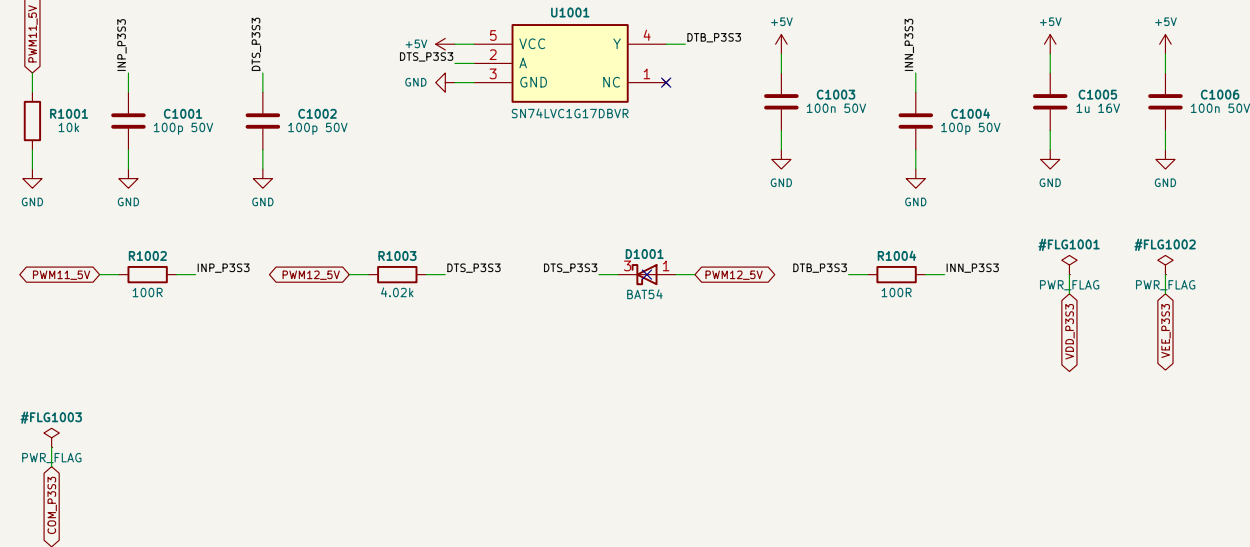


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG A: LAYOUT RULES

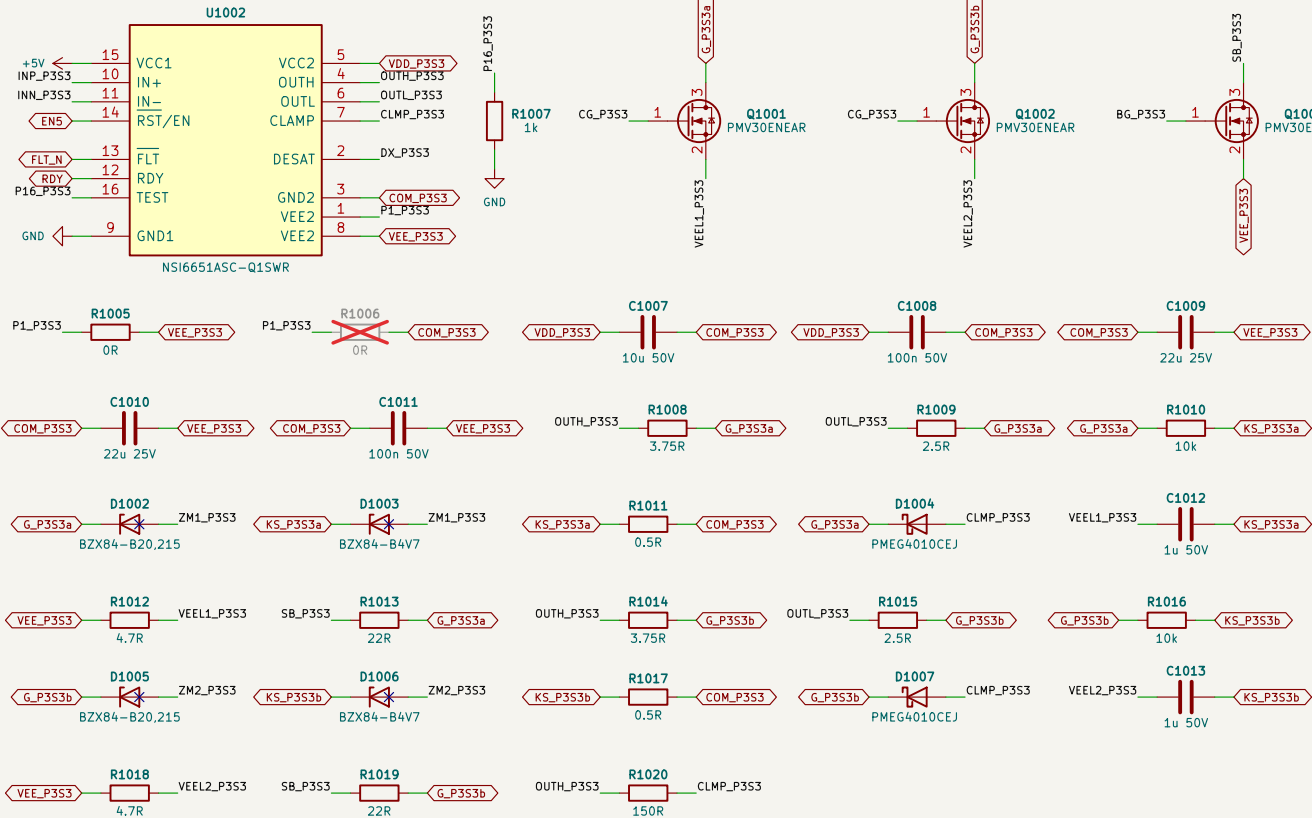
Gate and Kelvin of each device as coupled pairs of equal length (<=10%) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device; Clamp FET + C-VL at its device's gate-Kelvin pins, loop <= 1 nH; DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22); Miller hold (nospice, gdrv); die +1.41 V vs VGS(in) min 1.9k V at 175 C - BENCH TEST; Driver input side and Bias primary decoupled to GND locally (CM current).

P3S3: LOGIC SIDE



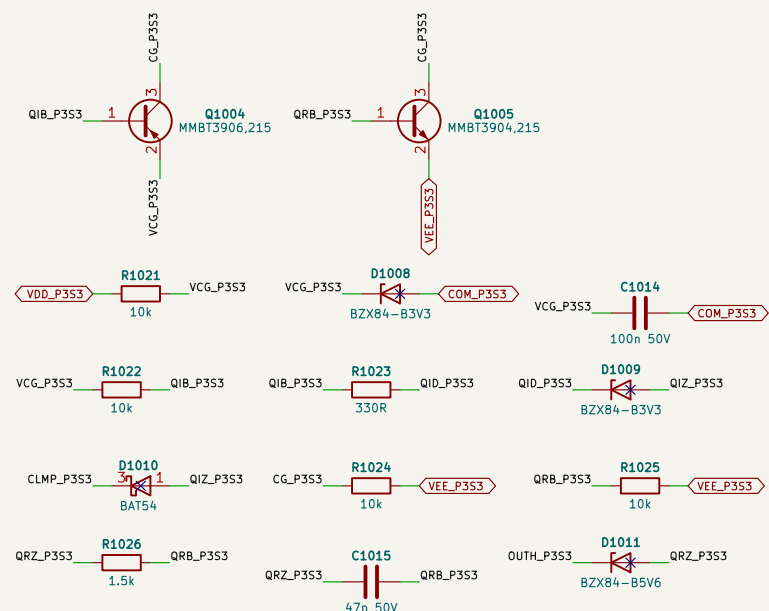
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



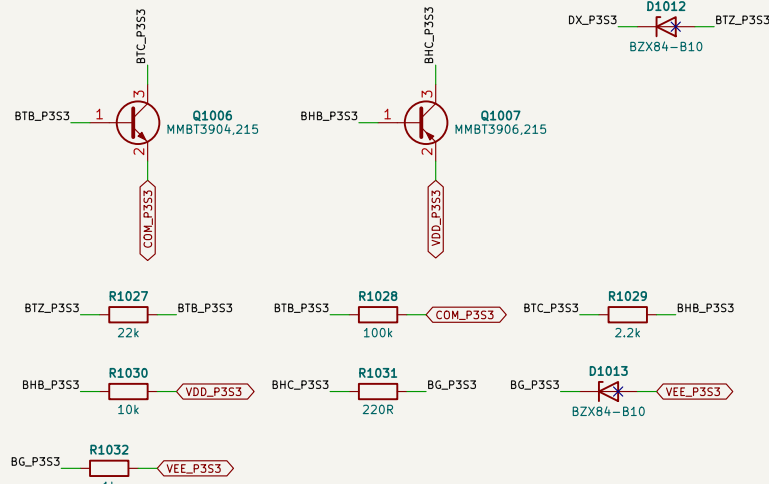
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



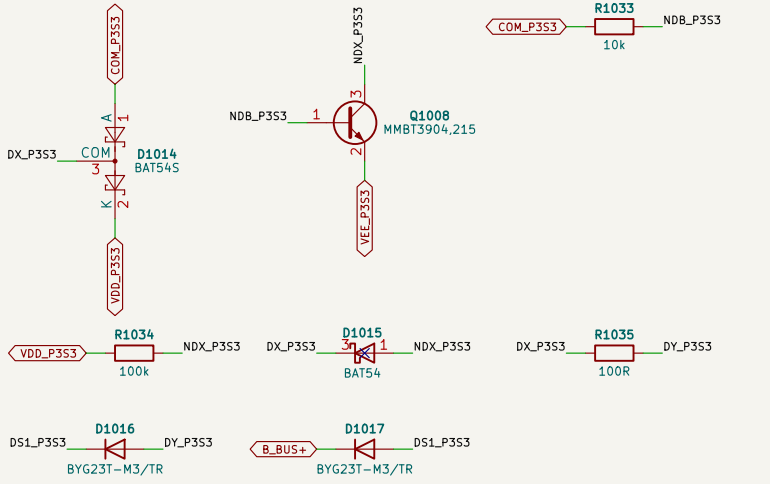
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



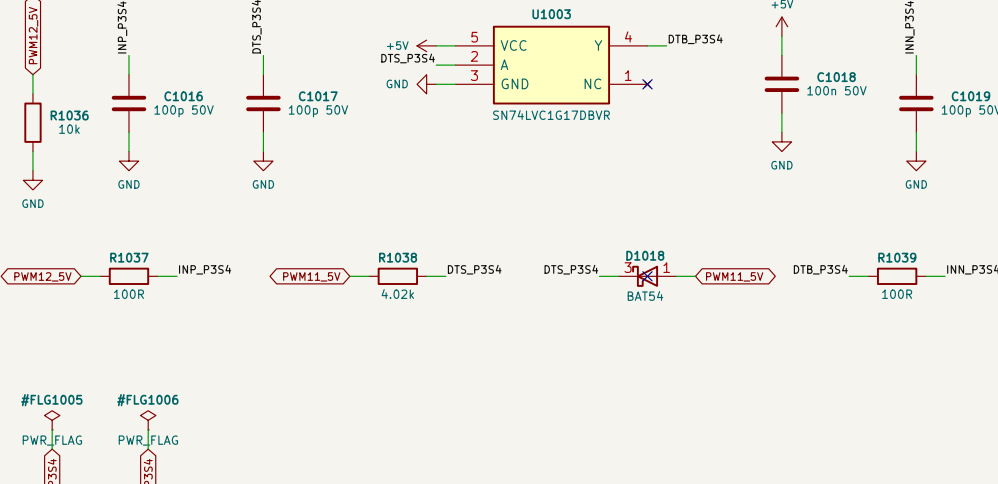
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P3S3: DESAT



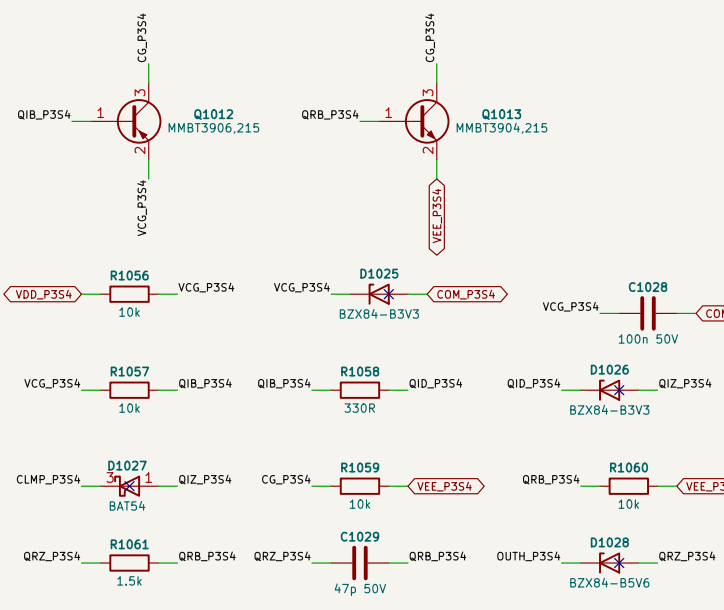
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

P3S4: LOGIC SIDE



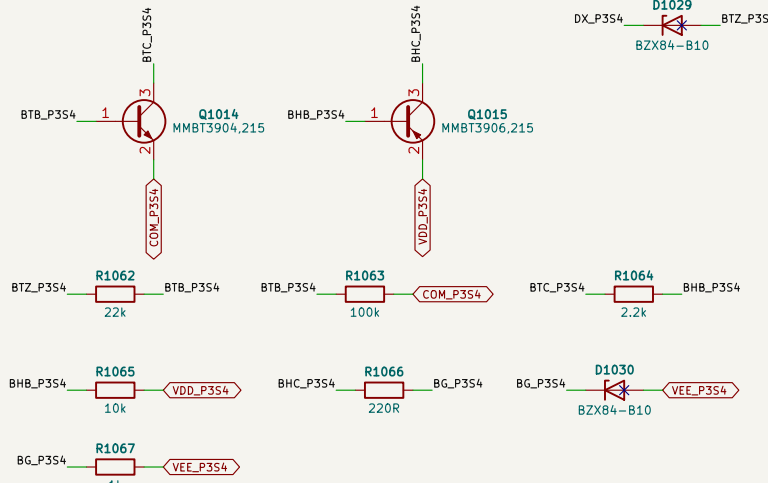
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S4: MILLER CLAMP CONTROL



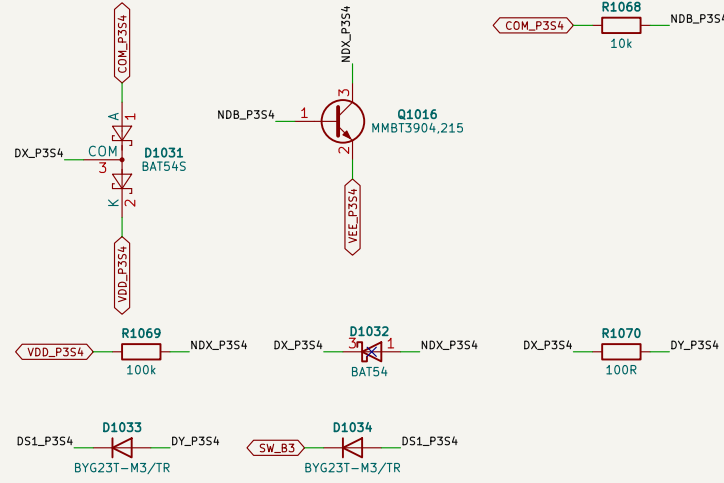
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P3S4: DESAT



DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

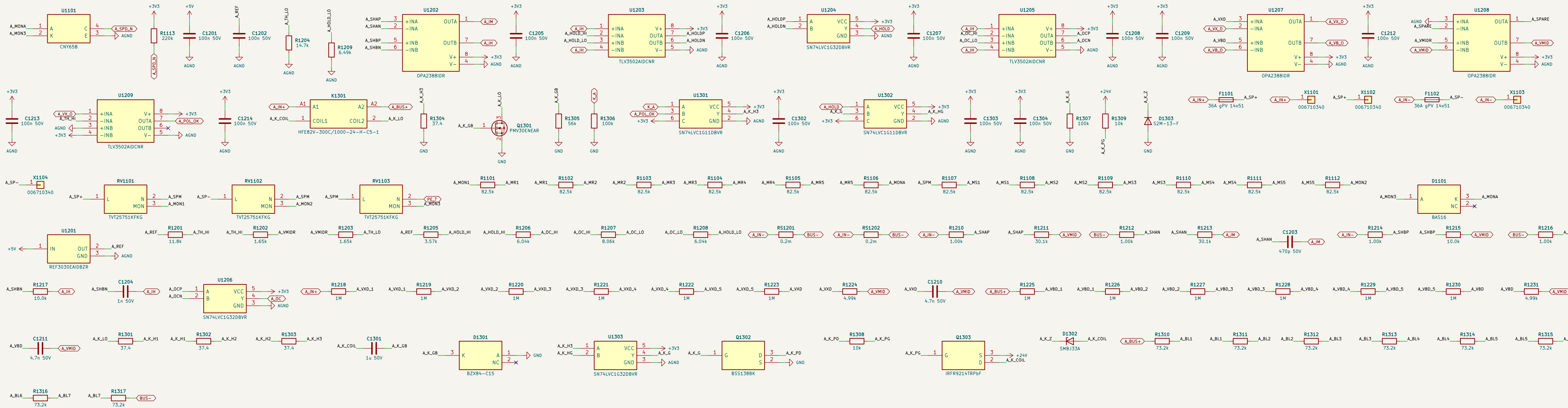
LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10%) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device; Clamp FET + C-VL at its device's gate-Kelvin pins, loop <= 1 nH; DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22); Miller hold (naspice, gdr); die +1.41 V vs VGS(in) min 1.9k V at 175 C - BENCH TEST; Driver input side and Bias primary decoupled to GND locally (CM current).

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_IN+ / A_IN-, bank A_BUS+. No fuse, no precharge; array Isc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV125751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/-M, MOV2 +/-M, MOV3 M/PE,I. Up.eff 3793 V pole-PE and pole-pole at In 3 kA (calc. <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); Imax 25 kA. Branch fuses ETI CH14x21 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line); RB1 = 1 at ms; above 200 V = varistor network open -> CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 13
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /12 Port A sensing and interlocks/
File: 12_leanA_ctrl.kicad_sch

Title: **PV-PWR power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 13/23

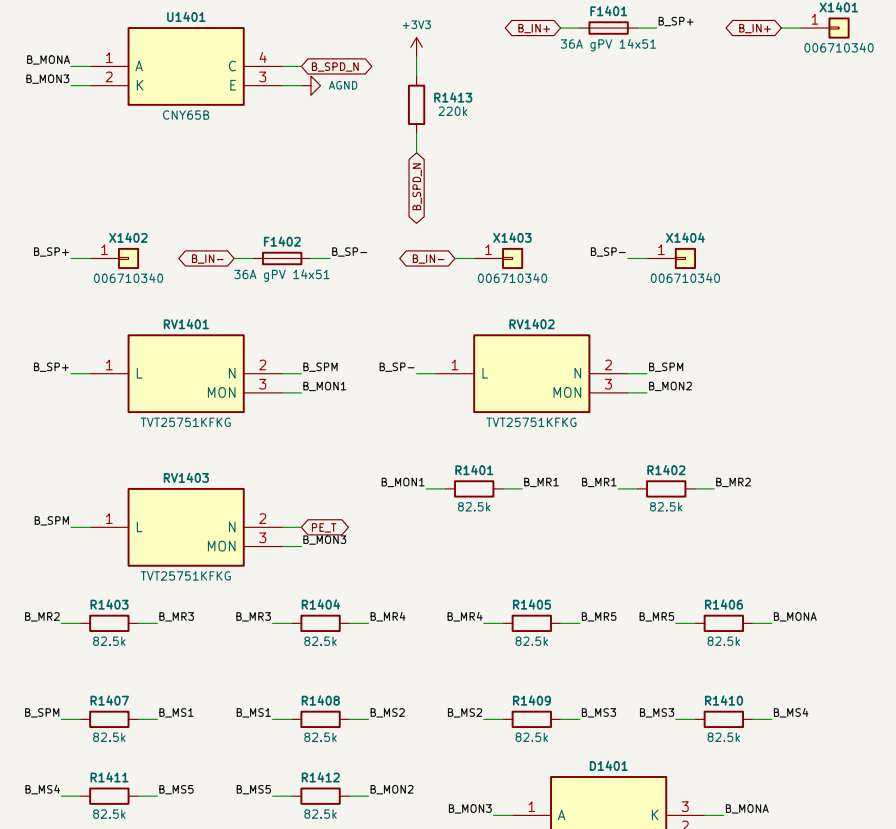
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 14 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /13 Port A coil drives/ File: 13_leanA_drive.kicad_sch		
Title: PV-PWR power board – Port A coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 14/23	

PORT B (BATTERY SOURCE), D-044 LEAN PORT

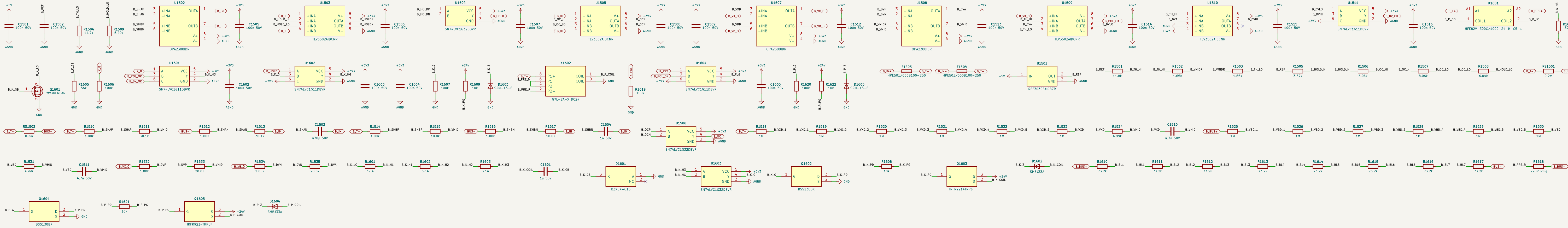
Terminal side R_MN+ / R_MN- bank R_BUS+/- HPE501 250 A at in both poles (breaks 1.25-50 A at only).
Contactor opens <= 0.97 A, held above (port_Lapc time). Installation: upstream circuiters 987-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV25751 = 3 (605 VDC, thermal time + monitor load); MOV1 + PM MOV2 - 7M MOV3 W/PEL1 Use at 3795 V pole-PE and post-pole at 3 VA (GND, <= 4.0 VV => reinforced 5 VV / basic 4 V, D-032); max 25 mA Branch
From ESI CHANSEL GPF 50.5 50 mA (D/F 2 ms); Monitor loop through 500 Tera and all time (data) -> CHANSEL
-> R81 (sum on the ID line); R82 = 1 at rest above 200 V = varistor network open -> CTR1 inhibit/stop.

AR FUSES (BATTERY-TYPE PORT)



HPE501/0008100-250 in both poles breaks 1.25-50 A at only (partial range);
the part QC trip and the contactor keep it out of 250 A-1.25 A.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /14 Port B sensing and interlocks/
File: 14_leanB_ctrl.kicad_sch

Title: **PV-PWR power board – Port B sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

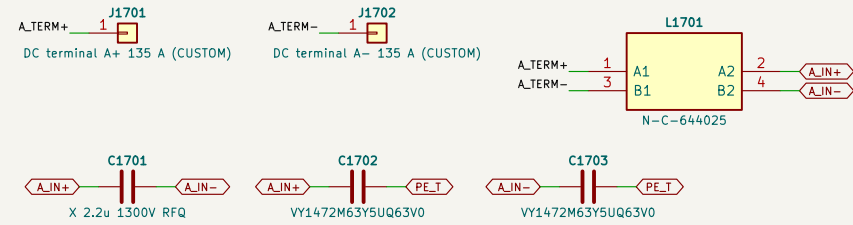
KiCad E.D.A. 10.0.6

Id: 16/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

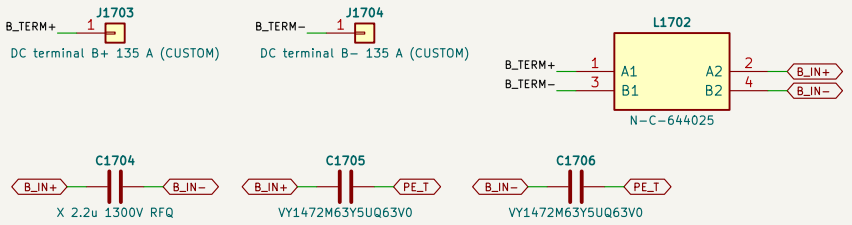
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 17 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Port B coil drives/ File: 15_leanB_drive.kicad_sch		
Title: PV-PWR power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 17/23	

PORT A TERMINALS AND EMI FILTER



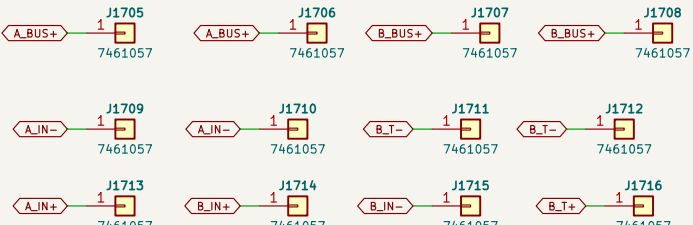
PV: chassis terminals A_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



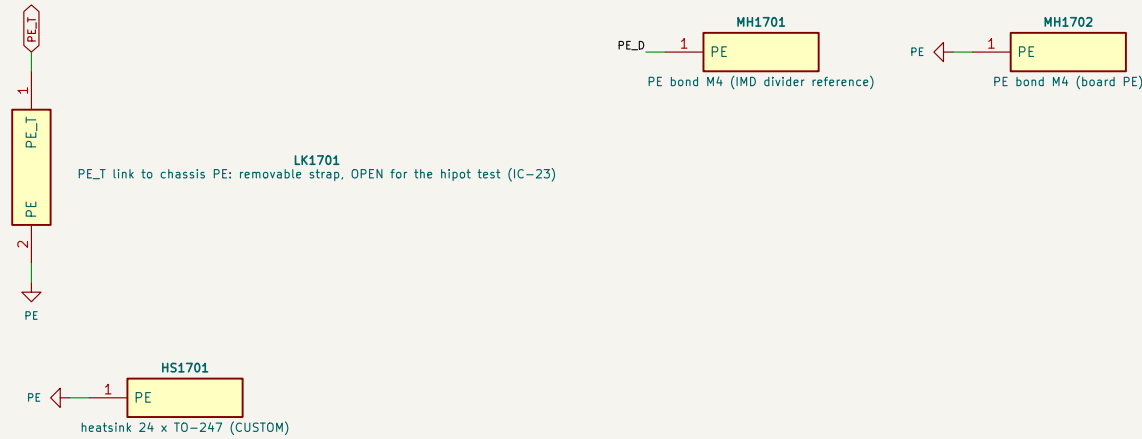
battery: chassis terminals B_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



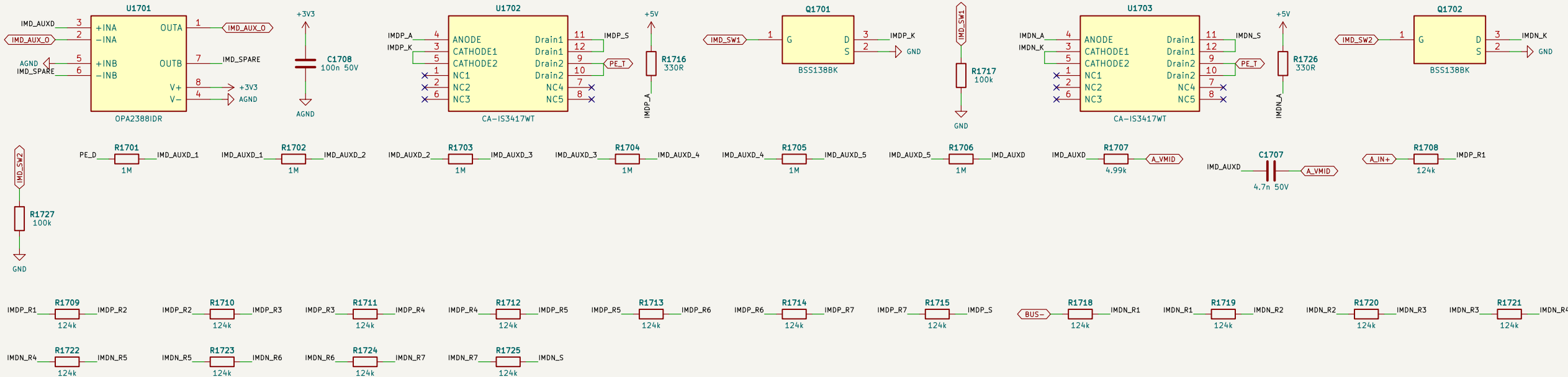
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected);
VPE = PE divider against BUS- (VMID of port A).

Not bench-validated: Values read from sim/out/*.json; checks: design_check()

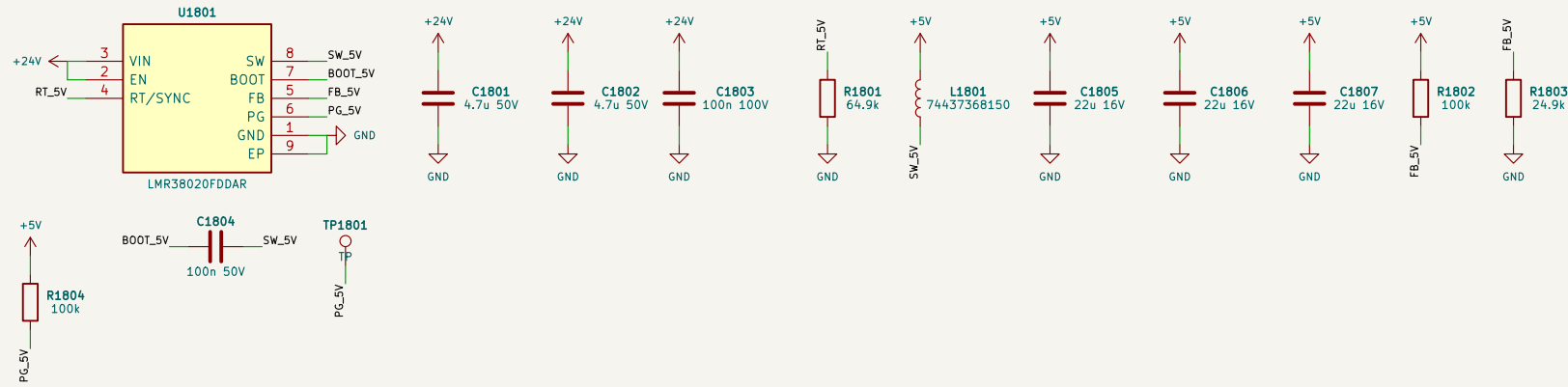
Sheet 18
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Terminals, EMI filter, insulation test/
File: 15_terminals_imd.kicad_sch

Title: PV-PWR power board - Terminals, EMI filter, insulation test

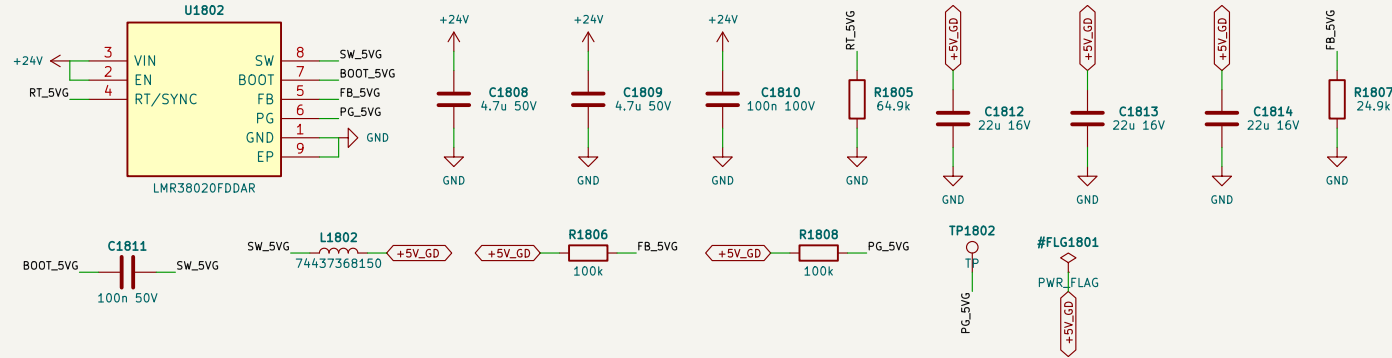
Size: A3 Date: 2026-10-05 Rev: A2
KiCad E.D.A. 10.0.6 Id: 18/23

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



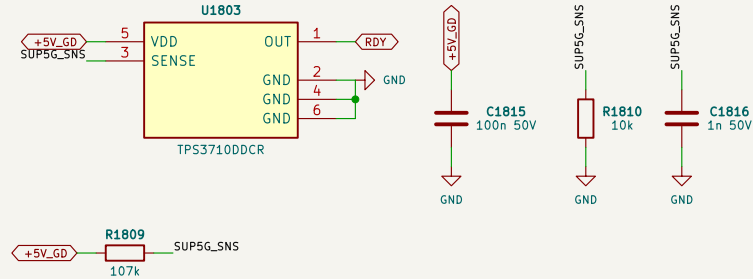
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



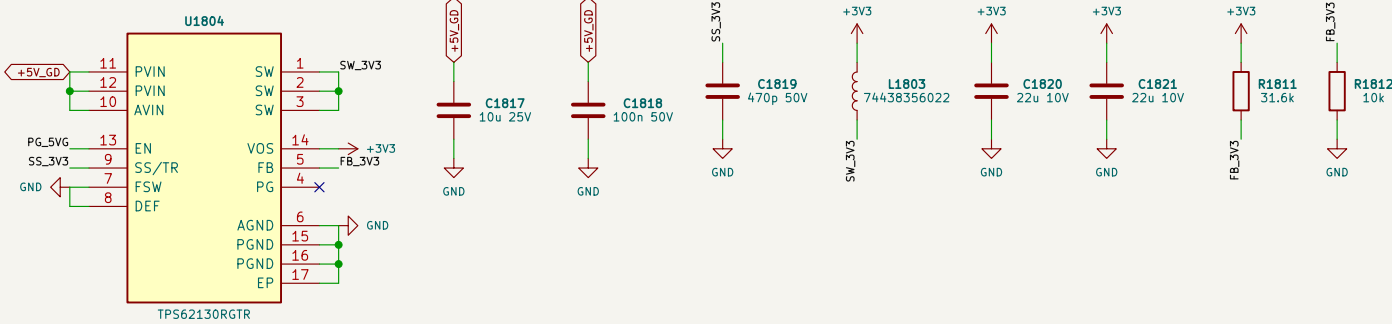
gate bias of phases 3 only, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



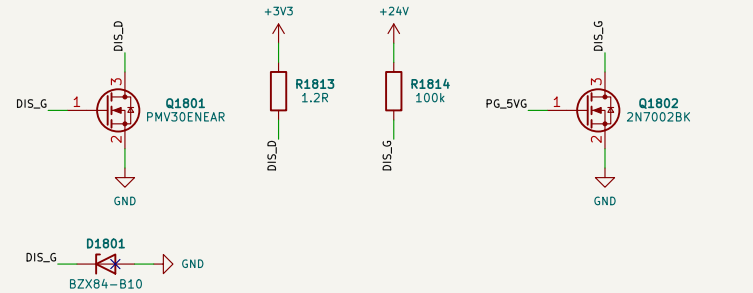
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (>= 3.20 V for the control board's reference), SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 21.4-30.1 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 19

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Live 5 V / 3.3 V, SELV connector/

File: 16_supplies.kicad_sch

Title: PV-PWR power board - Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

Id: 19/23

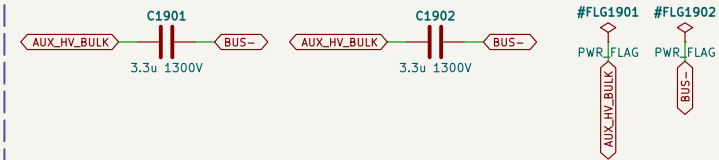


2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting



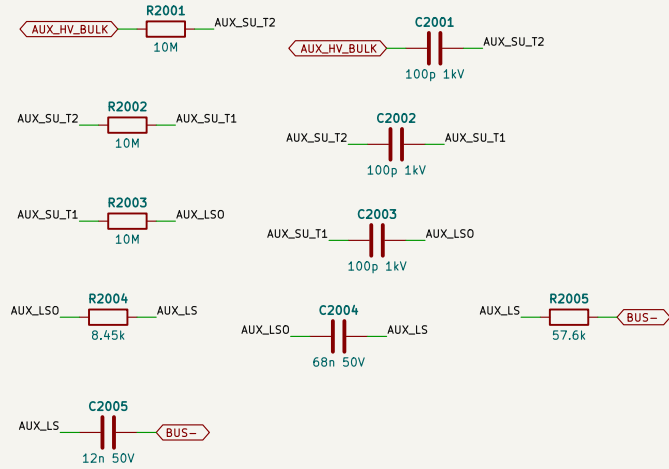
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



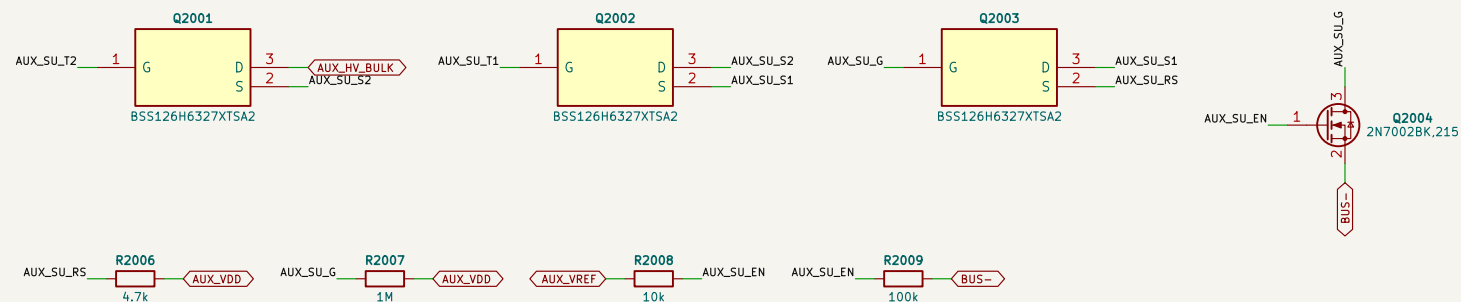
Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

COMPENSATED TAP STRING



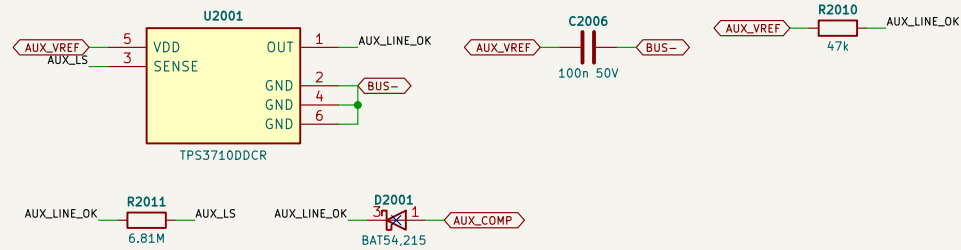
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



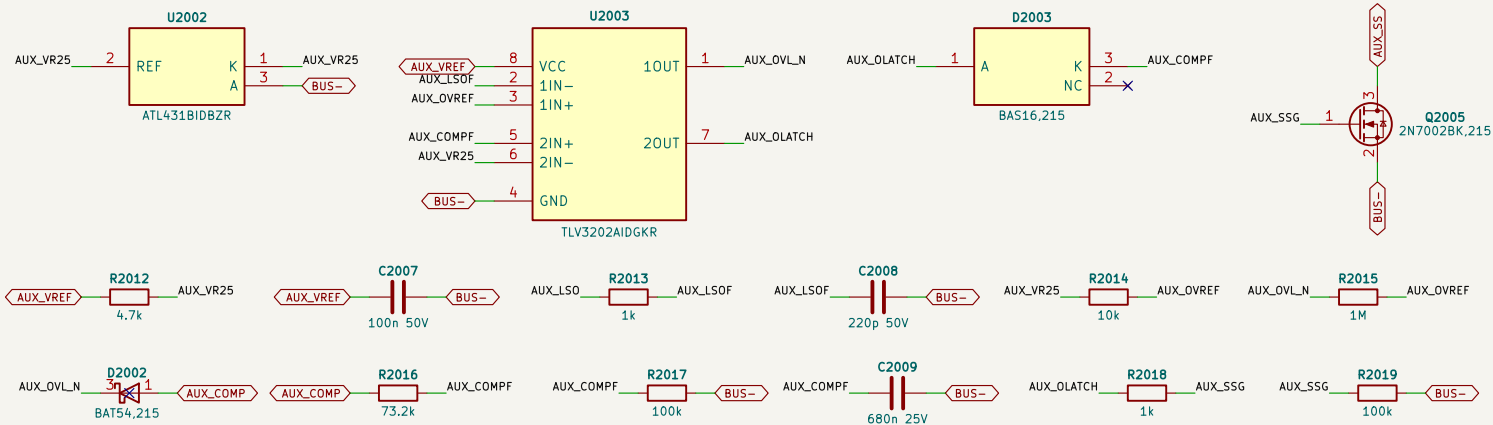
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



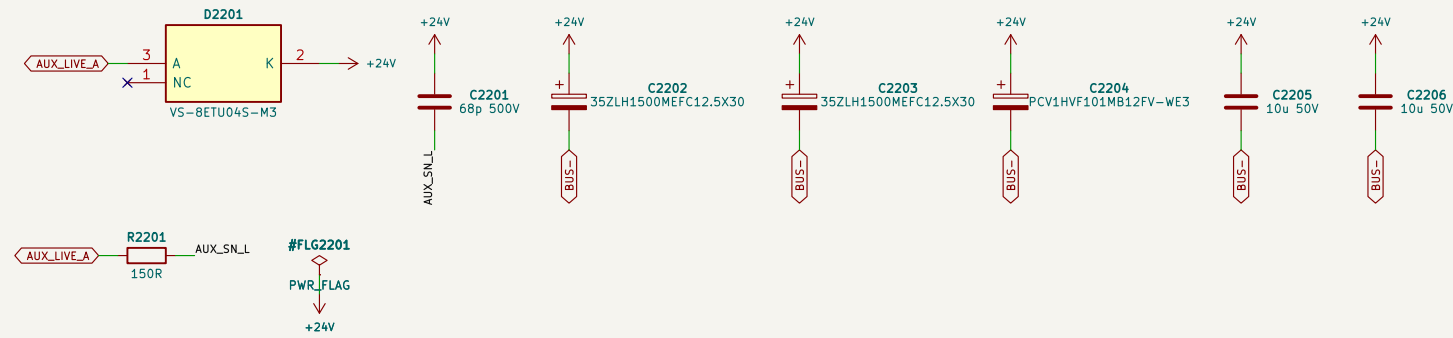
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



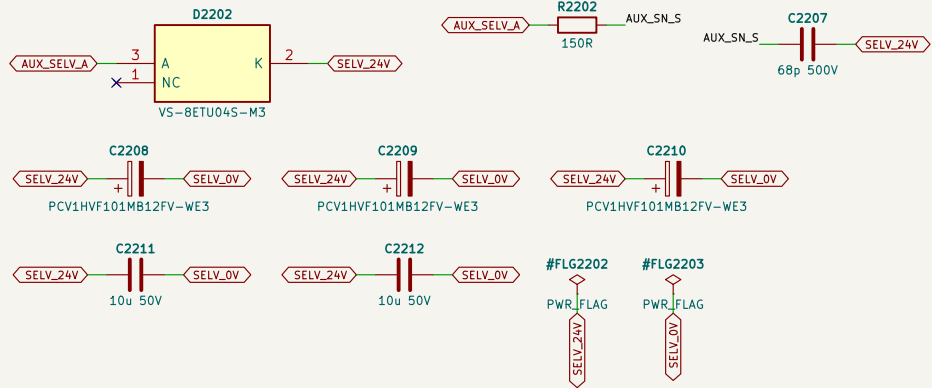
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 113 / 175 / 278 W (no trip at the 112.2 W 1 s peak); timer 39–100 ms -> latch -> SS crowbar -> hiccup.

LIVE 24 V RECTIFIER, HOLD-UP



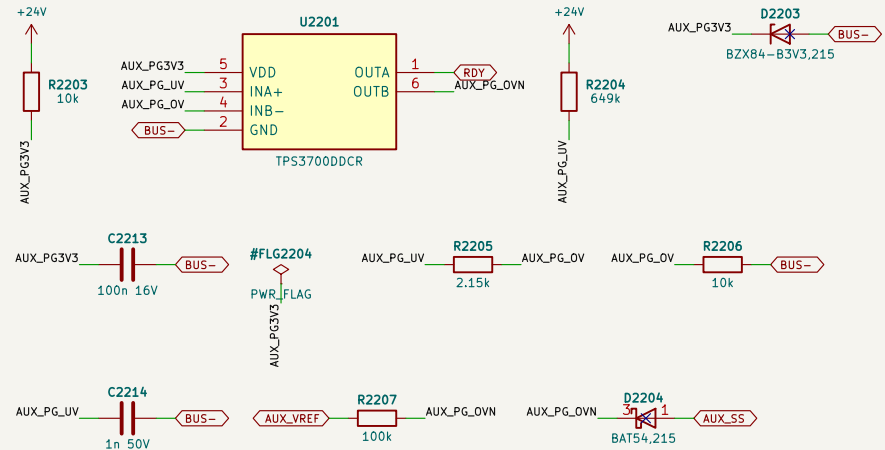
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02-21.81 V (status), OV rises 26.13-26.76 V (SS pulled low)